Evolutionary Dynamics of Corporate State Structures in Fifty States: Analyzing How Corporate Tax Impact Business Strategies and Economic Growth Using Advanced Data Analytics

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Abstract

The exploration of state-wise tax structures holds paramount significance due to its implications for various stakeholders, governments, businesses, and citizens. The comprehension of tax policy distinctions among different states plays a pivotal role in strategic decision-making processes in geographical expansion, investment decisions, and cost optimization strategies. This thesis aims to analyze different tax structures for businesses in various states and how they transitioned throughout the years using sophisticated data analytics tools such as R studio and Microsoft Excel. This thesis will additionally undertake a comparative analysis between CNBC's assessment of the optimal state for entrepreneurial activity and the corresponding ranking of corporate tax rates. This investigation aims to discern the potential correlation between these two metrics, shedding light on their interrelationship within the broader context of business environment evaluation These tools will be instrumental in answering pivotal questions such as:

- How have varying state tax structures transitioned and transformed throughout the years? And how does it impact business decision-making processes, particularly in terms of geographical expansions?
- How do these differences impact businesses' operational costs and profit margins?
- How can organizations optimize their operations considering the varying tax structures?
- What correlations exist between the varying tax structures and economic growth in different states?

With the support of these analytical tools, the study seeks to unveil correlations, patterns, and significant factors in varying tax structures that made a consequential impact on businesses in different states. Ultimately, this study aims to offer valuable guidance and strategic perspectives to businesses, policymakers, and other stakeholders in navigating tax-specific frameworks.

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Statement of the Problem and Aims of the Study

Statement of the problem:

The exploration of state-wise tax structures carries pivotal significance to its stakeholders, governments, businesses, and citizens. A deeper understanding of tax policies across diverse states can be crucial for decision-making processes, particularly in geographical expansions, investment strategies, and cost optimization initiatives for businesses. Even with the ever-evolving tax structures over time, their significance towards businesses is still crucial. To review, this study seeks to address the following matters:

1. How have tax structures transitioned from 2018 to 2023?
2. How have these transformations impacted decision-makers in terms of business growth?
3. What strategies can organizations adopt to optimize their operations?
4. What correlation exists between variation in tax structures and in-state economic growth?

CNBC has introduced a methodology for assessing the optimal state for commencing business endeavors. However, this methodological framework emerged notably in 2021, postdating the economic repercussions of the Covid-19 pandemic that beset the United States in 2020. This incident raises leading questions towards:

1. Does the delineated methodology sufficiently capture the comprehensive economic landscape of the assessed states?
2. To what extend this methodological approach can reliably predict the future economic conditions within each state?

Aims of the study:

This study aims to comprehensively analyze the dynamics of state-wise tax structures and their evolution throughout the years in various states, while utilizing sophisticated data analytics tools, such as Microsoft Excel and R studio. The goal of this study is as follows:

1. Explore and understand the historical changes and transformations within tax structures over the years in various states.
2. How much impact do the transitions factor on decision-making processes with business expansion and operational costs?
3. Develop and recommend the best strategies to optimize business costs in various states.
4. Identify and analyze correlations between tax structures and broader economic growth with the various states.

In conjunction with CNBC's method for identifying the best states to start businesses between 2021 and 2023, this study will employ Microsoft Excel to conduct an analytical analysis. The aim is to predict and forecast the rankings of specific states, thereby enhancing our understanding of future entrepreneurial prospects.

Hypothesis:

In the ever-changing economic landscape, states across the nation have undergone significant shifts in their tax policies. These changes play a crucial role in shaping the decisions of businesses, particularly concerning expansion and cost management. This study aims to use advanced data analytics tools to investigate these tax policies and their correlations with economic growth in different states.

By utilizing sophisticated tools like R Studio and Microsoft Excel, this research endeavors to delve deep into the complex relationships between tax structures and the economic performance of individual states. Through rigorous computational analysis and robust statistical modeling, the study seeks to uncover patterns, reveal correlations, and derive meaningful insights from the intricate interplay of state-specific tax policies and economic dynamics.

Furthermore, this thorough analytical approach seeks to go beyond mere observation, transitioning into a platform for strategic decision-making. The ultimate goal of these analytical efforts is to provide strategic recommendations grounded in empirical evidence, aiding businesses in navigating the complex operational landscape shaped by the diverse tax structures found across states. This scholarly endeavor aims to offer actionable insights that are both nuanced and contextually relevant, contributing to the broader discussion on effective business strategies amidst varying state tax environments.

In the analytical framework employed herein, the intricate tax systems of each state necessitate a comprehensive examination. To streamline computation, the determination of average tax rates becomes essential, serving as a key metric for subsequent evaluation. Methodologically, states are systematically arranged based on their computed average tax ratings, facilitating discerning analysis across the spectrum of numerical values from smallest to largest.

This systematic arrangement clarifies, over the temporal trajectory spanning 2018-2023, the states characterized by the most modest average corporate tax rates. The meticulous application of this methodological approach endeavors to provide a nuanced understanding of the variances in corporate tax impositions across diverse states during the specified period.

The assessment of optimal states for commencing business operations is executed through a meticulous methodology devised by CNBC, underpinned by a comprehensive consideration of 10 distinct variables. These variables, comprising cost of doing business, infrastructure, life, health, and inclusion, workforce, economy, business friendliness, access to capital, technology and innovation, education, and cost of living, collectively contribute to the overall rating, with each variable assigned a specific weightage as a percentage within the comprehensive 100% framework.

In the expansive assessment, Virginia claimed the top spot for business initiation in 2021, eventually ceding its position to North Carolina for the subsequent years of 2022-2023. As an

additional layer of scrutiny, we delve into the average tax rates spanning 2021-2023, comparing them with the states recognized by CNBC as the prime locations for commencing business endeavors. This comparative exploration offers a perceptive lens through which we can discern the interrelationship between the preferred states for business initiation and the corresponding fluctuations in their average tax rates during this specified timeframe.

New York City is widely acknowledged as a global financial hub, earning it the moniker "the bank of the world." Consequently, living expenses in the city are notably higher compared to other regions in the United States. This includes elevated taxation, particularly burdensome for businesses. The city's strong financial infrastructure, coupled with the presence of numerous corporate headquarters, is expected to drive prices upward across various sectors. Consequently, New York City may rank unfavorably due to its reputation for high living costs**.**

Evolutionary Dynamics of State Structures in Various States: Analyzing Impact on Business Strategies and Economic Growth Using Advanced Data Analytics

1. Tax Rating

In today's intricate economic landscape, the in-depth knowledge of state-wise tax structures is crucial in shaping the trajectory of businesses and governments. The intricate web of tax structures is woven across various states playing an essential role in steering business strategic decisions and operational dynamics to maintain or accelerate growth that influences economic development. Amid the economic regulations, the imperative historical evolution can impact state tax structures that emerge as a cornerstone for essential informative decision-making and transformative strategies. This thesis endeavors to commence an in-depth exploration of tax structures using sophisticated data analytics tools such as R studio and Microsoft Excel as a platform to explore history, comprehend the present, and project future potentials.

Over the years, corporate tax structures in each state have undergone dynamic transformations, influencing businesses in multifaceted ways. This study also dwells into the intricacies of these transitions, seeking answers to crucial questions that shape the business landscape, such as evolutionary corporate state tax structures, operational impacts, optimizing strategies, and correlation with economic growth. Anticipated findings from this research will contribute to informing decision-making processes while aiding in the development of effective strategies for navigating diverse tax strategies.

Annually, individual states prescribe distinct tax brackets accompanied by corresponding tax rates. The dynamic nature of these tax brackets reflects the responsiveness of certain states to the evolving economic landscape, with adjustments being made to accommodate such changes. Conversely, there are states where these tax structures remain immutable. Notably, in states such as Washington, a departure from the conventional corporate tax framework is observed, with taxation predicated upon gross receipts. This discussion explains how this unique tax system works. An example is set below:

Alaska	0.00%	>	$0	2018
	2.00%	>	$25,000	2018
	3.00%	>	$49,000	2018
	4.00%	>	$74,000	2018
	5.00%	>	$99,000	2018
	6.00%	>	$124,000	2018
	7.00%	>	$148,000	2018
	8.00%	>	$173,000	2018
	9.00%	>	$198,000	2018

	9.40%	>	$222,000	2018

(Scarboro, 2018)

1.1 Tax Brackets and Regulations

The tabular representation above delineates the tax brackets applicable to Alaska in the fiscal year 2018. Notably, it specifies that corporations falling within the income range of $1 to $24,000 will be subject to a 2% corporate tax rate. Correspondingly, entities realizing an income within the bracket of $25,000 to $48,000 will incur a 3% corporate tax, and this progressive structure persists throughout subsequent income tiers.

Some states do not have tax brackets and will imply consistent corporate tax regardless on their profit. Listed below are some of the states:

California	8.84%	>	$0
Colorado	4.63%	>	$0
Connecticut (b)	8.25%	>	$0
Delaware (c)	8.70%	>	$0
Florida	5.50%	>	$0
Georgia	6.00%	>	$0

(Scarboro, 2018)

Tax specifications exhibit distinct characteristics across various states, as illustrated in the table. Noteworthy examples include 'Connecticut' marked with the identifier (b) and 'Delaware' designated by (c), indicating nuanced tax calculation methodologies inherent to these states. The ensuing enumeration delineates the specific tax specifications applicable during the fiscal year 2018:

(a) 2018 -In Arkansas, corporations with net income over $100,000 pay 5.4 percent on the first $100,000 plus 6.5 percent on the excess over $100,000.
(b) 2018 - Rate includes a 10% surtax, which effectively increases the rate from 7.5% to 8.25%. Surtax is required by businesses with at least $100 million annual gross income
(c) 2018 - Nevada, Ohio, Texas, and Washington do not have a corporate income tax but do have a gross receipts tax with rates not strictly comparable to corporate income tax rates. See Table 18 for more information. Delaware and Virginia have gross receipts taxes in addition to corporate income taxes
(d) 2018 - Illinois' rate includes two separate corporate income taxes, one at a 7.0% rate and one at a 2.5% rate.
(e) 2018 - The tax rate in Indiana will decrease to 5.75% on July 1, 2018.
(f) 2018 - Corporations with entire net income greater than $100,000 pay 9% on all taxable income, companies with entire net income greater than $50,000 and less than or equal to

$100,000 pay 7.5% on all taxable income, and companies with entire net income less than or equal to $50,000 pay 6.5% on all taxable income. Note: In addition to regular income taxes, many states impose other taxes on corporations such as gross receipts taxes and franchise taxes. Some states also impose an alternative minimum tax and special rates on financial institutions. (Scarboro, 2018).

This thesis engages in a meticulous examination of tax rating dynamics, scrutinizing the transformative trends spanning the years 2018 through 2023. The assessment encompasses the delineation of tax brackets, serving as a visual aid to elucidate the allocation of corporate taxes across diverse entities and the consequential taxation of profits. To discern optimal states for corporate investment, a comprehensive analysis unfolds through the annual computation of average tax rates for each state, facilitated by the use of Microsoft Excel as a computational too.

1.2 Zero Tax Rating States

Upon rigorous examination and computational scrutiny, the findings suggest a consistent pattern over the temporal span from 2018 to 2023, wherein four specific states consistently maintain their status as the foremost locales for commencing business activities, characterized by the absence of corporate taxation, registering at a rate of 0%.

2018			2019		
1	Nevada	0	1	Nevada	0
2	Ohio	0	2	Ohio	0
3	Texas	0	3	Texas	0
4	Washington	0	4	Washington	0

2020			2021		
1	Nevada	0	1	Nevada	0
2	Ohio	0	2	Ohio	0
3	Texas	0	3	Texas	0
4	Washington	0	4	Washington	0

2022			2023		
1	Nevada	0	1	Nevada	0
2	Ohio	0	2	Ohio	0
3	Texas	0	3	Texas	0
4	Washington	0	4	Washington	0

In the year 2018, North Carolina attained a commendable status by being ranked among the top five optimal states for initiating business activities. Subsequent to this achievement,

however, its standing experienced a decline in 2019, superseded by North Dakota. Notwithstanding this setback, North Carolina demonstrated resilience, reclaiming its position in the succeeding years from 2020 to 2023, notably fortified by the implementation of a judicious 2.5% average corporate tax rate.

2018			2019		
1	Nevada	0	1	Nevada	0
2	Ohio	0	2	Ohio	0
3	Texas	0	3	Texas	0
4	Washington	0	4	Washington	0
5	North Carolina	3.00%	5	North Dakota	2.49%

2020			2021		
1	Nevada	0	1	Nevada	0
2	Ohio	0	2	Ohio	0
3	Texas	0	3	Texas	0
4	Washington	0	4	Washington	0
5	North Carolina	2.50%	5	North Carolina	2.50%

2022			2023		
1	Nevada	0	1	Nevada	0
2	Ohio	0	2	Ohio	0
3	Texas	0	3	Texas	0
4	Washington	0	4	Washington	0
5	North Carolina	2.50%	5	North Carolina	2.50%

1.3 States with Lowest Corporate Tax Ratings

The following list outlines the five states with the highest corporate tax rates:

2018			2019		
45	Iowa	9.00%	45	Delaware	8.70%
46	New Jersey	9.00%	46	California	8.84%
47	Illinois	9.50%	47	Illinois	9.50%
48	Minnesota	9.80%	48	Minnesota	9.80%
49	Pennsylvania	9.99%	49	Pennsylvania	9.99%

2020			2021		
45	California	8.84%	45	Delaware	8.70%
46	Iowa	9%	46	California	8.84%

47	Illinois	9.50%	47	Illinois	9.50%
48	Minnesota	9.80%	48	Minnesota	9.80%
49	Pennsylvania	9.99%	49	Pennsylvania	9.99%

2022			2023		
45	Delaware	8.70%	45	Delaware	8.70%
46	California	8.84%	46	California	8.84%
47	Illinois.	9.50%	47	Pennsylvania	8.99%
48	Minnesota	9.80%	48	Illinois	9.50%
49	Pennsylvania	9.99%	49	Minnesota	9.80%

Exploring the quintet of states identified as the most challenging for entrepreneurial endeavors, as per the computed averages, Pennsylvania emerges as a consistent contender within this undesirable cohort. A high corporate tax rating, persisting at 9.99% from 2018 to 2022, establishes Pennsylvania as a prominent member of this ranking. Notably, a slight amelioration is observed in 2023, as the state ascends to the 47th position with a reduced corporate tax rate of 8.99%, surpassing Minnesota in this trajectory.

Minnesota, perpetually situated at the lowest point of the ranking from 2018 to 2023, maintains an unwavering corporate tax rating of 9.8%. However, the alteration in Pennsylvania's tax policy in 2023 precipitates a shift, positioning Minnesota as the ultimate state in terms of unfavorable conditions for commencing a business.

The penultimate position, assigned to Pennsylvania as number 49, acknowledges an outlier state that deviates from the conventional corporate tax framework. South Dakota and Wyoming, occupying the 50th and 51st position, is distinctive in its absence of corporate state taxes, thus rendering any percentage or average calculation moot. It is imperative to note that this distinct taxation paradigm in South Dakota and Nevada are not inherently disadvantageous; rather, it signifies an exemption from corporate taxes, distinguishing it from its counterparts.

The inclusion of a 51st entity in this analysis is necessitated by the presence of an additional outlier, Washington D.C. Although Washington D.C. is not designated as a state, but rather as a federal district, its unique status originates from the congressional establishment of a federal district in 1790, placing it under the jurisdiction of the U.S. Congress. Functioning as both a state and concurrently executing municipal and county responsibilities, Washington D.C. operates with a distinctive administrative framework. Residents of the district are mandated to fulfill civic obligations, including tax payments, voting, participation in jury duty, and military service (Statehood).

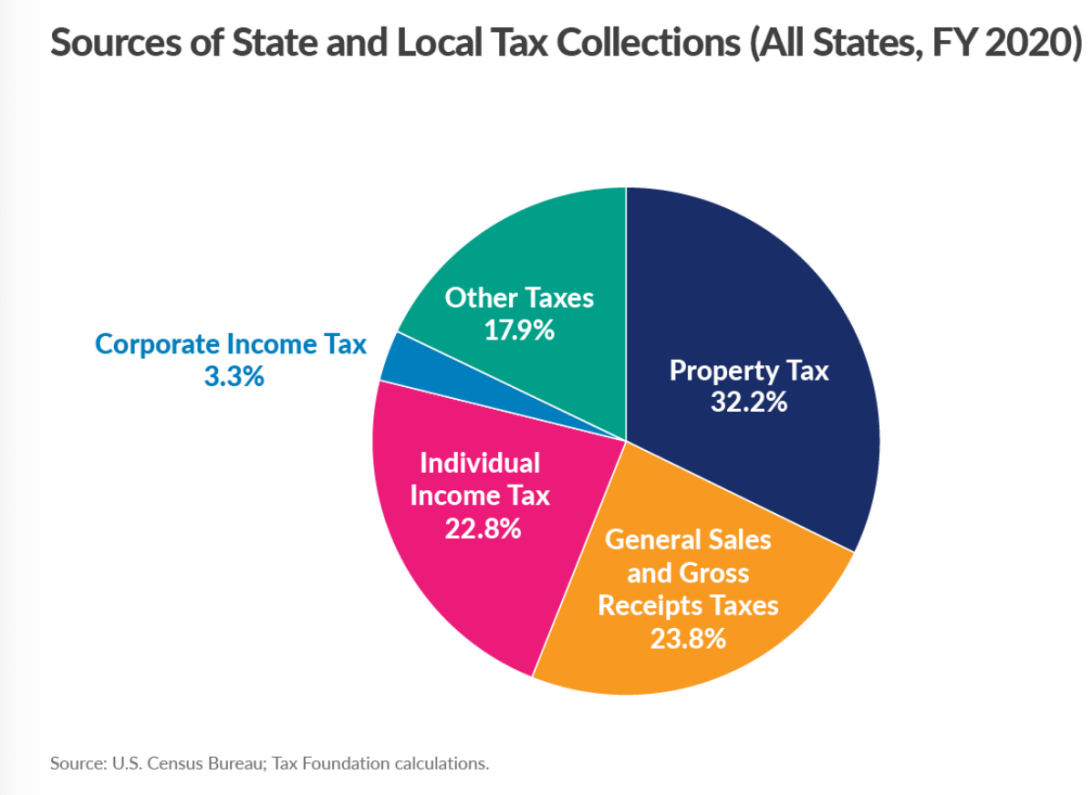
1.4 Absence of Corporate Tax Rates

Wyoming distinguishes itself by its absence of individual income tax and corporate tax impositions. The state is characterized by a state sales tax rate of 4%, complemented by a maximum local sales tax rate of 2%, culminating in an average combined state and local sales tax rate of 5.36%. Notably, this fiscal landscape propels Wyoming to secure the foremost position in the 2023 State Business Tax Climate Index, denoting its advantageous position in fostering a favorable business tax environment (Foundation).

Per official records, corporate taxes in the fiscal year 2020 constituted a modest 3.3% of the overall state and local tax revenue. Notably, corporate taxes emerge as one of the more volatile revenue streams for states. Conversely, property taxes assume a preeminent role, serving as the primary source of tax collections at the local level, contributing significantly, up to 72.2%, to local tax revenue in fiscal year 2020. Sales tax, generating approximately 32.3% of the state tax revenue in the same fiscal year, stands as a consequential revenue source for states, as it does across the nation. It is imperative to underscore that sales tax also plays a pivotal role in constituting a substantial portion of the local government revenue stream (Katherine Loughead, Unpacking the State and Local Tax Toolkit: Sources of State and Local Tax Collections (FY 2020), 2022).

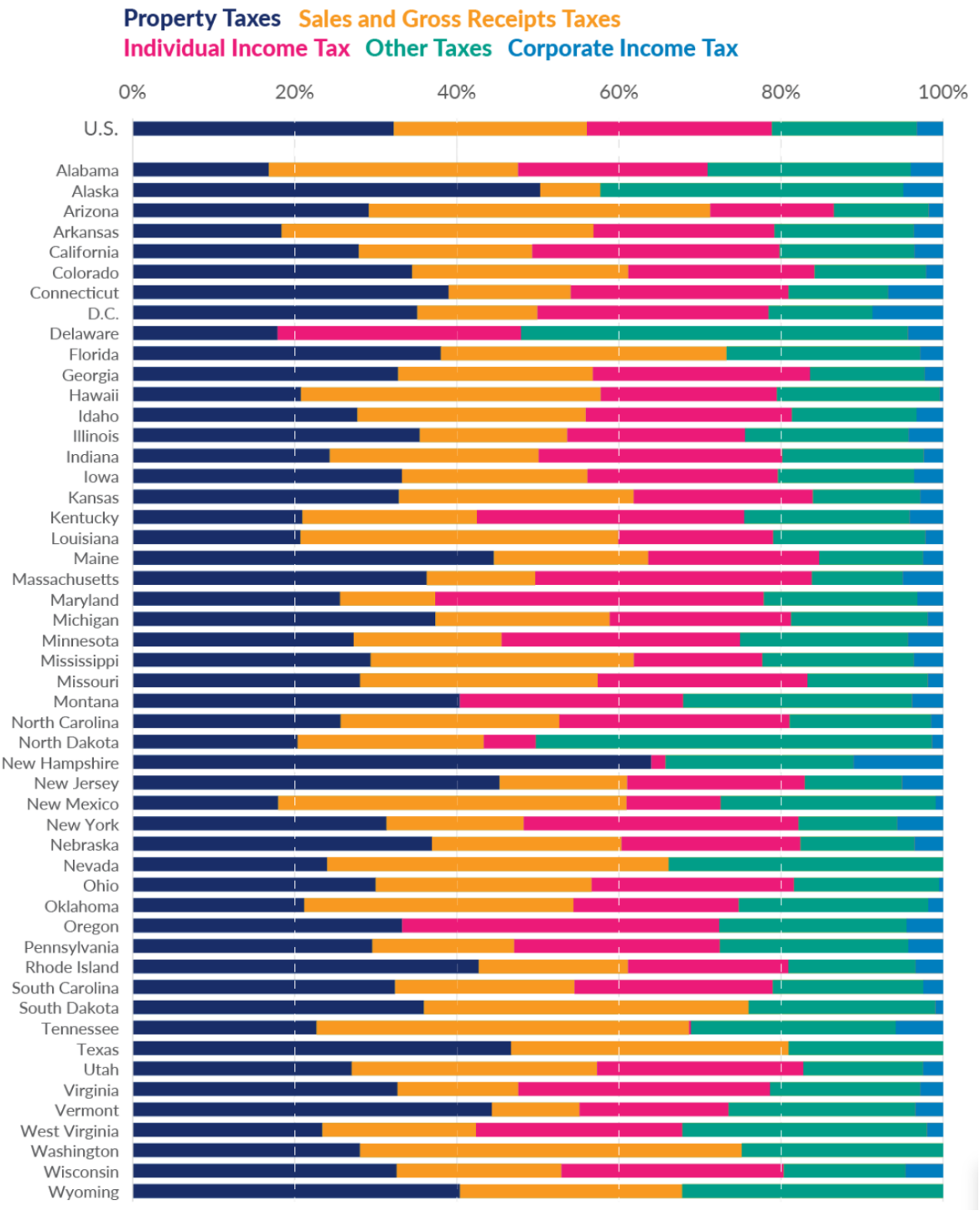
This intricate interplay of revenue sources elucidates how states, particularly those with low or no corporate tax, adeptly navigate and compensate through reliance on alternative taxation mechanisms, ensuring a continued capacity for robust local tax collections (Katherine Loughead, Unpacking the State and Local Tax Toolkit: Sources of State and Local Tax Collections (FY 2020),

2022).



Analogously, South Dakota adheres to a tax framework completely lacking of individual income tax and corporate income tax, featuring a 4.5% state sales tax rate, a maximum local sales tax rate of 4.5%, and an average combined state and local sales tax rate of 6.4%. This comprehensive fiscal structure positions South Dakota favorably, securing the 2nd rank in the 2023 State Tax Climate Index. (Katherine Loughead, Unpacking the State and Local Tax Toolkit: Sources of State and Local Tax Collections (FY 2020), 2022).

In consonance with its fiscal model, South Dakota relies extensively on property taxes as its primary tax resource, contributing substantially, at approximately 72.2%, to local tax revenue in fiscal year 2020. Echoing a parallel dynamic observed in Wyoming, corporate taxes in South Dakota represent a relatively modest 3.3% contribution to the overall local tax revenue in the fiscal year 2020. A visual depiction elucidating the distinct tax collection methodologies employed by each state is provided in the accompanying illustration below, offering insights into their respective localized tax revenue generation mechanisms (Katherine Loughead, Unpacking the State and Local Tax Toolkit: Sources of State and Local Tax Collections (FY 2020), 2022).



Note: "Other taxes" includes selective sales taxes (excise taxes), motor vehicle license taxes and fees, business license taxes, inheritance and estate taxes, recordation fees, and transfer taxes, among others. While most states' gross receipts taxes are reported in Census's "general sales" tax category, Delaware's GRT revenue is categorized as "other selective sales" tax revenue and included in our "Other Taxes" category. Oregon's Corporate Activity Tax (CAT), a gross receipts

tax, took effect January 1, 2020. CAT revenue is sent to the Fund for Student Success rather than the General Fund. FY 2020 included six months of CAT collections, but CAT collections do not appear to show up in the FY 2020 Census data for "general revenue from own sources."

Source: U.S. Census Bureau; Tax Foundation calculations. (Katherine Loughead, Unpacking the State and Local Tax Toolkit: Sources of State and Local Tax Collections (FY 2020), 2022)

2. The State Business Tax Climate Index to Evaluate Tax Systems

The State Business Tax Climate Index serves as a crucial instrument facilitating the evaluation of tax systems by business leaders, government policymakers, and taxpayers. This index is strategically designed to elucidate the efficacy with which states structure their tax systems, offering a comprehensive benchmarking mechanism. Functioning as a nuanced roadmap, the Index not only benchmarks the current state of tax structures but also delineates pathways for improvement, enabling stakeholders to make informed decisions and fostering a continuous discourse on optimizing the fiscal landscapes of respective states (Jared Walczak, 2023).

2.1 Framework

The Tax Foundation State Business Tax Climate Business Index is a hierarchical framework compromising five distinct components (Jared Walczak, 2023):

- Individual income tax
- Sales tax
- Corporate income tax
- Property tax
- Unemployment insurance tax

Each component of the Tax Foundation State Business Tax Climate Index is meticulously scored on a scale ranging from 0 (indicating the least favorable) to 10 (indicating the most favorable). These components are systematically organized to represent major domains within state taxation, each encompassing a multitude of variables. In aggregate, the measured variables outlined in the report collectively contribute to a total of 125 points (Jared Walczak, 2023).

States exhibiting a higher standard deviation are characterized by substantial competitive advantages, embodying a dynamic reminiscent of a push and pull theory. Illustratively, Delaware, with its zero-tax rate on sales, serves as an exemplar of this phenomenon. The absence of sales tax in Delaware renders it an attractive destination, magnetizing businesses and consumers from across states. As alternative states augment their sales tax rates, Delaware's allure intensifies, accentuating the differential attractiveness and competitiveness across state tax landscapes.

Notably, the equitable weighting of these components does not adhere to a uniform distribution; rather, the assignment of weights is contingent upon the variability of scores across

the 50 states relative to the mean. The calculation involves the determination of the standard deviation for each component, and the subsequent creation of weighted measures based on this variability analysis conducted by Jared. The weighted components are as following (Jared Walczak, 2023):

Tax Rate	Tax Type
29.80%	Individual Income Tax
23.30%	Sales Tax
20.90%	Corporate Tax
14.90%	Property Tax
11.10%	Unemployment Insurance Tax

2.2 Variables

In contradistinction to the Index, it is imperative to underscore the discernible sensitivity inherent in the overall ranking to minor alterations in the rates of individual components. A nuanced observation reveals that even slight modifications in the rates of each component can exert a disproportionate influence on the comprehensive recalibration of the overarching ranking. According to Jared, within each component there are two equally weighted variables: scalar variables and dummy variables. "A scalar variable is one that can have any value between 0-10. If a subindex is composed only of scalar variables, then they are weighted equally. A dummy variable is one that has only a value of 0 or 1" (Jared Walczak, 2023). Consider the case of indexing tax brackets for inflation – states may choose to either incorporate this practice or refrain from doing so. The integration of scalar (numeric) and dummy (categorical) variables within a subindex can introduce challenges, particularly as the extreme valuation of a dummy variable might exert undue influence on the subindex results. In response to this potential distortion, the Index adopts a standardized approach, assigning an 80 percent weight to scalar variables and a 20 percent weight to dummy variables. This balanced weighting mechanism aims to mitigate the impact of extreme values and foster a more robust and nuanced evaluation of the subindex.

Consider a hypothetical scenario involving two states, State A and State B, both partaking in the Index evaluation process. One of the components under scrutiny is the indexing of tax brackets for inflation:

- State A adopts the practice of indexing its tax brackets for inflation, thus reflecting a scalar variable of 1 for this specific component.
- State B opts not to index brackets for inflation, leading to a scalar variable of 0 for the same component.

Now, considering only the scalar variables without addressing the influence of dummy variables, the overall impact of this specific component on the subindex might be disproportionately skewed by the extreme value of the dummy variable (0 or 1). To counteract this

potential distortion, the Index employs a standardized approach. Scalar variables, in this case, would carry an 80 percent weight, providing a nuanced reflection of the state's tax policy.

Simultaneously, dummy variables, with a 20 percent weight, ensure that categorical choices don't disproportionately dominate the overall subindex. This methodology aims to foster a fair and balanced evaluation of each state's tax policies within the Index framework. Assigning a numerical score of "0" does not denote the absolute nadir in terms of viability for commencing a business. Instead, it signifies the least favorable positioning within the cohort of the 50 states under scrutiny. It is imperative to underscore that this score of "0" is contextualized within the specific subset of states considered, thereby establishing a relative evaluation framework. This methodological refinement is crucial for fostering a nuanced understanding, elucidating that the score reflects the state's comparative standing among the 50 entities assessed rather than an absolute qualitative judgment.

The intricacies inherent in tax rate variations across states necessitate a nuanced analytical approach, wherein the absolute index proves insufficient to furnish the requisite precision for discerning the optimal tax landscape conducive to business enterprises. The resultant challenge compels pragmatic business proprietors to grapple with the arduous task of ascertaining the most favorable tax environments among states. Notably, against the backdrop of the substantial global investment surge observed in 2001, governments discerned the pivotal role played by states' corporate tax policies in shaping the decisions of foreign investors.

In an ideal scenario, states refraining from dependence on both corporate income tax and gross receipt tax emerge as candidates for achieving an optimal neutral tax framework with respect to business income. Such states, characterized by the absence of reliance on these taxation modalities, stand poised to attain a theoretical pinnacle in achieving a neutral and equitable system. The attainment of this hypothetical zenith is correspondingly reflected in the assignment of a perfect score, signifying the states' alignment with the ideal of a tax environment conducive to business income. (Cammenga, 2012).

2.3 Single-System vs Multiple Brackets

South Dakota and Wyoming stand as unique entities among the states, abstaining from levying corporate income tax or gross receipt tax, earning them an automatic perfect score of 10. The ensuing comparison involves the remaining 48 states against each other, introducing an evaluative framework hinged on distinct tax brackets, each assigned a corresponding rate.

The highest acclaim is reserved for a single-rate system featuring a solitary bracket applicable to the entirety of taxable income, epitomizing a straightforward and equitable approach. A favorable disposition is also extended to systems where the top rate is activated at a lower income threshold, aligning with the principles of a flat tax. Conversely, states boasting multiple brackets find themselves at the less advantageous end of the spectrum.

The number of brackets further influences corporate income taxation dynamics. As corporate income escalates and attains its zenith within a bracket, the tax rate undergoes alterations. This fluctuation bears economic consequences, acting as a potential deterrent for companies as income levels rise. In this vein, Alaska emerges with the least favorable standing, characterized by a 10-bracket system. Following suit, Arkansas presents a 5-bracket system, with New Jersey and Maine in tow, both navigating a 4-bracket system, marking them within the less preferable spectrum of the evaluative framework (Jared Walczak, 2023).

Alaska	0.00%	>	$0	2018
	2.00%	>	$25,000	2018
	3.00%	>	$49,000	2018
	4.00%	>	$74,000	2018
	5.00%	>	$99,000	2018
	6.00%	>	$124,000	2018
	7.00%	>	$148,000	2018
	8.00%	>	$173,000	2018
	9.00%	>	$198,000	2018
	9.40%	>	$222,000	2018

2.4 Evaluation

The evaluative framework employs three weighted criteria to assess the competitiveness of each state's corporate tax base rate, each criterion bearing equal significance. These criteria encompass the accessibility of specific credits, deductions, and exemptions. States subject to gross receipt taxes are assessed based on the taxation of gross receipts concerning employee compensation costs and the cost of goods sold. According to the IRS, gross receipts are the total amounts the organizations received from all sources during its annual accounting period, without subtracting any costs or expenses (IRS, 2023).

Net operating losses (NOL) are subject to a deliberate tax regime wherein they are exclusively taxed on profits. However, acquiring a comprehensive annual snapshot of a corporation's genuine profit proves inherently challenging. The deduction for net profit loss (NOL) plays a pivotal role in ensuring that companies are subjected to taxation on their income, allowing for the averaging of profits over time.

Under the aegis of the Tax Cuts and Jobs Act, the federal government extends the allowance for losses to be carried indefinitely. However, companies are restricted to reducing taxable income by a mere 80% annually. In agreement with this federal stance, states adopting a gross receipts tax model are treated equivalently to having no NOL provisions for carrybacks or carryforwards.

Noteworthy is the provision allowing NOL deductions to extend up to 20 years and retroactively back to a period spanning 3 years.

Certain states impose restrictions on the quantum of carryback and carryforward net operating loss (NOL) amounts, a parameter accorded due weightage in the Index. States imposing limitations on these capacities are reel gated to lower rankings within the evaluative framework. Notably, Idaho and Montana exhibit relative restraint by capping their carryback provisions at $100,000 and $500,000, respectively, despite their superior standing compared to counterparts not offering carryback provisions.

In the realm of carryforward losses, a more restricted landscape unfolds, with only New Hampshire, Illinois, and Pennsylvania enacting limitations on carryforward amounts. Illinois, in a recent legislative move, has instituted a cap of $100,000 on carryforward provisions spanning the years 2021-2024. The presence of these limitations contributes to the suboptimal performance of these states in this evaluative variable, as delineated within the Index (IRS, 2023).

3. How to Rate Top States to Start a Business Methodology

Other methodologies exist to assess the viability of commencing a business in the United States, among which CNBC has undertaken a comprehensive evaluation of all 50 states since 2021. The evaluative framework employed by CNBC is rooted in a meticulous consideration of 10 overarching categories, namely the cost of doing business, infrastructure, life, health, and inclusion, workforce, economy, business friendliness, access to capital, technology and innovation, education, and cost of living (Staff C. , America’s Top States for Business 2021, 2021). This comprehensive approach ensures a nuanced examination of various facets integral to the business environment.

The quantitative underpinning of this assessment is articulated through a structured methodology, wherein each state is subject to scoring with a maximum achievable tally of 2,500 points. Noteworthy is the reliance on empirical data sourced from diverse outlets, eschewing subjective opinions in favor of objective metrics. CNBC's discerning approach is emblematic of a commitment to factual accuracy and reliability.

According to the outcomes derived from CNBC's methodological rigor in 2021, the states of Virginia, North Carolina, Utah, Texas, and Tennessee emerged as the preeminent states, securing positions within the top five for commencing business endeavors (Staff C. , America’s Top States for Business 2021, 2021). These findings epitomize the substantive nature of CNBC's evaluations, grounded in empirical data and contributing valuable insights for prospective business ventures.

North Carolina has notably emerged as the preeminent state for commencing business operations during the period spanning 2022-2023. In the specific context of 2022, North Carolina has exhibited remarkable prowess, securing its status as the best state for starting a business. The

rankings, formulated through CNBC's meticulous methodologies, position North Carolina at lead, followed by a distinguish group consisting of Washington, Virginia, Colorado, and Texas within the top five states esteemed for business initiation.

A continued trajectory of excellence is observed in 2023, where North Carolina once again attains the coveted position as the premier state for commencing business activities. The discerning methodologies employed by CNBC delineate a nuanced landscape, with North Carolina leading the pack, accompanied by Virginia, Tennessee, Georgia, and Minnesota constituting the esteemed top five states for initiating business ventures. These rankings reflect the dynamic and competitive nature of states in fostering conducive environments for business inception, as discerned through the rigorous and methodical evaluations conducted by CNBC.

The variables under consideration are proposed and measured in the aftermath of the global pandemic, a period marked by an abrupt and comprehensive shutdown of the global economy. This period remained unaffected by ongoing social discruptions. These variables serve as proactive countermeasures devised to mitigate the risks associated with potential future economic disruptions. The genesis of these measurements is rooted in a pragmatic response to the imperatives of preventing unforeseen economic standstills, thereby engendering a more resilient and adaptive economic framework.

It is imperative to note that these measurements, embodying a forward-looking and anticipatory ethos, are exclusively available from the year 2021 onward. This focus on the time highlights their current relevance, designed to navigate and address the distinctive challenges precipitated by the global pandemic and to strengthen the economic landscape against analogous uncertainties in the future.

3.1 Variables

Governmental authorities have discerned that the conceptualization of the 10 categories encompasses a visionary reinterpretation of the Quality-of-Life domain (John, How we chose America’s Top States for Business in 2021 , 2021).

1. **Cost of Doing Business**

Within this evaluative schema, the cost of doing business, a crucial aspect, commands approximately 16% of the overall rating. The evaluative spectrum that is calibrated with the highest attainable rating equating to 100%, while the lowest point is marked at 50%. Consequently, the Cost of Doing Business segment is allocated a substantial 400 points, reflecting its integral role in the comprehensive evaluation.

The Cost of Doing Business category serves as a comprehensive measure, probing into the complexities of each state's business tax climate, the burden imposed by taxes, and the overall

infrastructural facilities. This encompassing assessment explores into the intricate interaction of incentives and tax breaks, strategically devised to alleviate the financial burdens on businesses. Of note is the emphasis on fostering development within disadvantaged communities, aligning with broader socio-economic considerations.

2. Infrastructure

The infrastructure category assumes a substantial weight, encompassing 375 points and contributing to 15% of the overall rating. The exigencies imposed by the global pandemic in 2020, notably the COVID-19 crisis, prompted a consequential reassessment of the supply chain, wherein the state's transportation system became a focal point of scrutiny. This comprehensive evaluation incorporates an intricate analysis of the value and volume of goods transported via air, waterways, railroads, and pavements. The efficacy of bridges, highways, and air travel further constitutes critical considerations. Accessibility to markets is gauged through a meticulous assessment of the population residing within a 500-mile radius of each state.

The evolving landscape induced by the pandemic instigated an expanded perspective on remote work, prompting an evaluation of the enhanced quality, availability, and pricing of broadband services within each state. Additionally, utility infrastructure assumes most important, with the condition of drinking water and wastewater systems, alongside the reliability of the electrical grid, held accountable within this evaluative framework.

Sustainability in the face of climate change emerges as a salient parameter, with due consideration to the availability of sites for expansion and development. This encompassing approach underscores the multifaceted nature of the infrastructure category, reflecting its pivotal role in determining a state's overall viability for businesses in a dynamically changing global landscape (John, How we chose America’s Top States for Business in 2021 , 2021).

3. Life, Health, and Inclusion

The life, health, and inclusion category, commanding a weight of 375 points and constituting 15% of the overall rating, emerges as a pivotal asset in the evaluative framework. The merge of technological advancement and heightened social consciousness, catalyzed by workforce shortages, has precipitated a transformative shift in corporate practices. Contemporary companies are now at the leading position of fostering more inclusive and hospitable working environments, underscoring the symbiotic relationship between technological evolution and societal values.

In the aftermath of the seismic events triggered by the 2020 COVID-19 pandemic, this category assumes augmented significance, earning distinction as one of the highest importance for considerations in 2021. The recalibration reflects an acute awareness of new dimensions in relevance to enhancing the quality of life, with sustained scrutiny on per capita crime rates, healthcare metrics, and environmental quality indicators. The imperative for robust safeguards

against discrimination and the protection of voting rights has been underscored through rigorous measures, aligning with legislative enactments as of June 1, 2021. This strategic alignment emphasizes a proactive approach to fostering an equitable and conducive societal and workplace milieu.

4. Workforce

The workforce category, wielding a weight of 325 points and representing 13% of the overall rating, assumes a salient role in the evaluative framework. The onset of the pandemic in 2020 precipitated widespread unemployment, with millions of Americans facing joblessness. Consequently, companies reported significant challenges in securing qualified workers, marking a pivotal juncture in workforce dynamics.

A nuanced assessment encompasses educational attainment metrics, considering the diminishing trends in educational achievements among the working-age and college-educated populations within each state. Worker productivity, gauged through economic output per job, serves as a vital metric, illustrating the intricate intersection between educational qualifications and economic contribution. Concurrently, considerations extend to the landscape of union memberships and the regulatory impact of right-to-work laws, both witnessing a decline, signifying a transformative period in labor relations.

Diversity and the availability of workers constitute additional dimensions integrated into the evaluative measurements, acknowledging the multifaceted nature of workforce considerations. This comprehensive approach underscores the evolving dynamics within the workforce category, capturing the complexities inherent in the post-pandemic employment landscape.

5. Economy

The economy category, allocated a maximum of 250 points and representing 10% of the overall rating, assumes a pivotal role in the evaluative schema. During periods of uncertainty, enterprises display a predilection for states characterized by fiscal stability and robust economic foundations. The assessment of economic and job growth serves as a barometer, illuminating the economic resilience of a state amid confusing times.

A rigorous evaluation of each state's fiscal condition encompasses multifaceted dimensions, including credit ratings and outlooks, as well as a comprehensive scrutiny of the overall budget. The latter incorporates spending patterns, revenue streams, and pension obligations, offering a holistic perspective on the fiscal health of a state. The health of the residential real estate market emerges as a critical variable, acknowledging the intricate interplay between economic robustness and the housing sector. Additionally, the prevalence of corporate headquarters within each state augments the evaluative framework, portraying the economic magnetism and strategic significance of each locale within the broader economic landscape. This comprehensive approach summarizes

the nuanced considerations integral to the economy category, reflecting its pivotal role in determining the economic viability of states for business enterprises.

6. Business Friendliness

The business friendliness category, contributing a total of 200 points and constituting 8% of the overall rating, assumes a pivotal role in the evaluative construct. In times of corporate crisis, businesses exhibit a proclivity for pathways marked by minimal impediments. This inclination towards the path of least resistance is intrinsically tied to the legal and regulatory frameworks within states, where simplicity and lack of complexity are deemed advantageous.

An intricate evaluation of each state's legal and regulatory landscape encompasses the nuanced realms of lawsuit and liability climates. The purview extends over areas such as labor and trade, encapsulating the comprehensive spectrum of bureaucratic intricacies. This discerning examination aims to distill the essence of business friendliness by scrutinizing the absence of impediments and complications within the legal and regulatory frameworks, thereby portraying the degree to which states facilitate a conducive environment for businesses navigating through challenging circumstances.

7. Access to Capital

The access to capital category, contributing a total of 175 points and encompassing 7% of the overall rating, assumes a critical role within the evaluative framework. The urgency imposed by the 2020 pandemic-induced economic shutdown compelled businesses to engage in resource mobilization, prompting states to proactively intervene in facilitating access to capital. Consequently, what was previously a peripheral consideration gained most importance, emerging as one of the pivotal factors to be considered in the post-pandemic landscape.

The heightened importance of access to capital is underscored by its ubiquitous presence in contemporary state marketing and rankings, marking a significant departure from historical trends. Variables within this category scrutinize the maximum amounts available through state-funded Covid-related loans and grants, leveraging federal aid programs such as the U.S. Small Business Administration Paycheck Protection Program (PPP) and Economic Injury Disaster Loans (EIDL) on a state-by-state basis. Beyond pandemic-related initiatives, the assessment extends to venture capital investments and bank loans, with an emphasis on the support provided to smaller businesses, further accentuating the multi-faceted considerations integral to the access to capital category.

8. Technology and Innovation

The technology and Innovation category, culminating in 175 points and constituting 7% of the overall rating, assumes a pivotal position within the evaluative framework. Competitive states are

acknowledged for their robust support of innovation, fostering an environment conducive to the cultivation of new ideas and the allocation of resources toward technological advancements. The evaluation in this category goes beyond typical measures, exploring various aspects that demonstrate a state's dedication to technological advancement and innovative progress.

States are subjected to scrutiny based on the outcomes of patents issued within their jurisdiction, providing a tangible metric for innovation. Furthermore, the allocation of grants in health, science, and agriculture serves as an additional barometer, delineating a state's investment in research and development across diverse domains. The evaluative framework broadens to encompass an exhaustive analysis of each state's technology ecosystem, carefully examining the vitality of its human capital, corporate entities, and investment landscape. This comprehensive approach summarizes the intricate interplay of factors that contribute to a state's technological and innovative vibrancy.

9. Education

The education category, commanding a total of 150 points and constituting 6% of the overall rating, assumes a crucial role in the evaluative construct. Given that most companies strategically seek a pool of educated workers, the emphasis on education extends beyond workforce considerations to encompass broader aspects such as providing an appealing environment for employees to raise families. The recruitment of new talent with higher educational backgrounds is particularly valued, aligning with the imperative for development and research partnerships.

The evaluative framework places significant importance on the abundance of higher education institutions within each state, recognizing their role in cultivating a skilled and knowledgeable workforce. The inclusion of K-12 education metrics broadens the scope, considering parameters such as test scores, class size, and expenditure, alongside provisions for lifelong learning opportunities. This multifaceted approach acknowledges the intricate interplay between various educational dimensions and their collective impact on the overall desirability of a state for businesses and their employees.

10. Cost of Living

The cost-of-living category, contributing a maximum of 75 points and constituting 3% of the overall rating, assumes a nuanced and strategic significance within the evaluative framework. In the aftermath of the economic downturn during the 2020 pandemic, anticipations of potential inflation have heightened. Against this backdrop, both companies and workers manifest an acute interest in states characterized by stable prices, where the daily cost of living remains affordable.

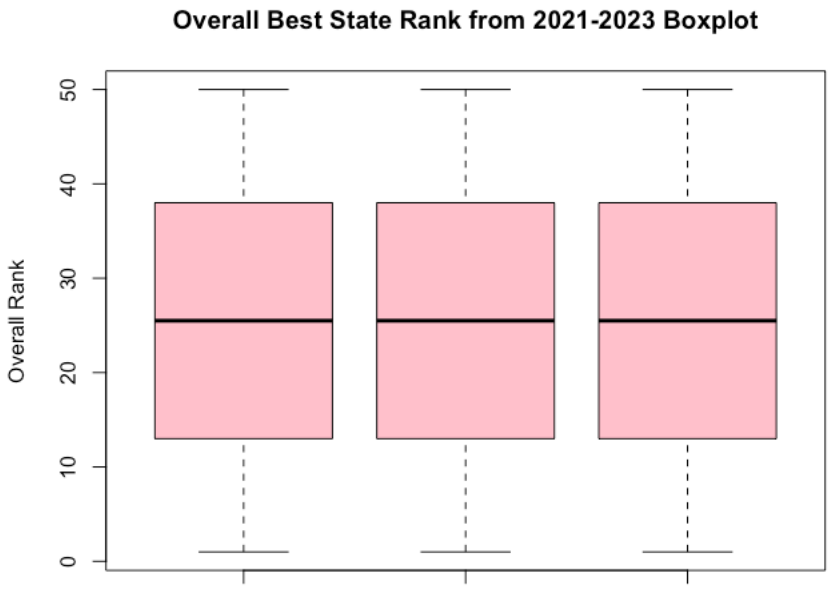
The intricate interrelation between the cost of living and the overall economic landscape is underscored by its profound impact on the cost of doing business. From housing to food and energy, the relative affordability of daily living directly influences the value of wages. States are

subjected to meticulous scrutiny through the lens of a cost index for basic items, providing a quantitative measure of the affordability and economic sustainability of the living conditions within each state. This holistic approach summarized the multifaceted considerations integral to the cost-of-living category and its consequential implications for the economic desirability of states for businesses and their workforce.

3.2 Overall Summary of Best State Rank from 2021-2023

Utilizing the statistical capabilities of R to extract key metrics such as the minimum, 25th percentile, median, 75th percentile, and maximum data offers investors valuable insights into determining the optimal state for their business endeavors, particularly for the year 2022. By analyzing the provided numerical data, investors can discern that any rank falling below 25.5 indicates subpar performance compared to the average. This crucial observation serves as a significant factor for investors to consider when selecting the most suitable state to establish their business operations based on the identified variables**.**

```
> fivenum(top22$`OVERALL RANK `)
[1]  1.0 13.0 25.5 38.0 50.0
> summary(top22$`OVERALL RANK `)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   1.00   13.25   25.50   25.44   37.50   50.00
```



Furthermore, the analysis of boxplot data across the years 2021-2023 reveals consistent maximum and minimum points, suggesting stability in these metrics. Additionally, given the presence of 50 consistently ranked states, it can be inferred that the median values remain unchanged over this period. This logical deduction underscores the reliability of median values across the three years, further enhancing their utility as indicative benchmarks for investors' decision-making processes.

As clarified in a 2021 article disseminated by CNBC, North Carolina occupies the second preeminent position, securing the coveted rank of number 1 as the optimal state for commencing business operations, coupled with boasting a robust and resilient economy. The gubernatorial administration, led by Governor Roy Cooper, has spearheaded strategic initiatives to fortify the economic landscape. Notably, the state clinched a monumental two-billion-dollar factory deal with VinFast, a Vietnamese electric vehicle manufacturer, indicative of a engaging strategy into transformative ventures.

A testament to North Carolina's economic magnetism, tech giant Apple has inked a consequential agreement to establish its inaugural East Coast hub within the state's Research Triangle region, facilitated by an incentives package totaling $846 million. This symbiotic collaboration underscores the state's commitment to fostering a cooperative milieu with significant investors, aimed at fostering positive outcomes that contribute to the steadfastness of the state's economic foundation.

Governor Cooper's governance philosophy prioritizes collaboration with major stakeholders to engender favorable economic conditions. This collaborative approach aligns with the state's steadfast commitment to cultivating high-paying employment opportunities, a facet prominently emphasized during negotiations with substantial investors. The article further underscores North Carolina's financial robustness, citing a pristine credit rating complemented by a notable economic growth rate of 6.7% in the year 2021, coupled with sustained job growth at an impressive 3.6%. This scholarly elucidation encapsulates the multifaceted elements contributing to North Carolina's ascendancy as the preeminent state for business initiation and economic vitality. (John, By putting partisanship aside, North Carolina ranks No. 1 in America’s Top States for Business, with the nation’s strongest economy, 2022).

However, as its overall results, in the year 2021, Virginia secured its position as the preeminent state in the nation, recognized for offering unparalleled wealth and talent. Although the specific calculations underlying the point allocation methodology were not disclosed, Virginia amassed a total of 1587 points out of a possible 2500. Notably, Virginia exhibited a remarkable recovery post-pandemic in 2020, boasting a robust population exceeding 8.5 million. The economic indicators further underscored the state's resilience, with a notable 6% GDP growth in the first quarter of 2021 and a commendably low unemployment rate of 4.5% (Staff C. , Virginia, 2021).

CATEGORY	2021 SCORE	2021 RANK
COST OF DOING BUSINESS	220	26
INFRASTRUCTURE	191	24
LIFE, HEALTH & INCLUSION	245	11
WORKFORCE	244	3
ECONOMY	158	13
BUSINESS FRIENDLINESS	137	11
ACCESS TO CAPITAL	140	9
TECHNOLOGY & INNOVATION	104	16
EDUCATION	119	2
COST OF LIVING	29	32
TOTAL	1587	1

3.3 Regression Analysis

Regression analysis serves as a pivotal statistical method to discern the variables that exert influence within a given context (Gallo, 2015). This analytical tool facilitates the clarification of relationships between dependent and independent variables, affording a comprehensive comprehension of the intricate interplay between variables and the way they mutually impact each other.

In the current context, a regression analysis will be conducted utilizing data extracted from CNBC's compilation of the "Top States to Start a Business" for the year 2021. Leveraging the Data Analysis tool embedded in Microsoft Excel and selecting Regression Analysis emerges as the methodological choice for its efficiency in generating insightful outcomes. The ensuing table presents the regression analysis results for the year of 2021:

Regression Statistics	
Multiple R	0.98276328
R Square	0.96582366
Adjusted R Square	0.95706049

Standard Error	3.02070286
Observations	50

ANOVA

	df	*SS*	*MS*	*F*	***Significance F***
Regression	10	10056.6388	1005.66388	110.214019	**2.17508E-25**
Residual	39	355.861186	9.12464579		
Total	49	10412.5			

The regression analysis outputs reveal insightful metrics for the model's performance. The multiple R, which stands at 0.98276328, signifies a robust correlation between the dependent variable (state rank) and the independent variables (ten categories determining the rank). This high correlation of 0.98 indicates a significant relationship.

Moving on to the ANOVA table, the F-statistic is 110.214019, with an extremely low p-value (2.17508E-25), signifying strong evidence to reject the null hypothesis. This emphasizes the significance of the model in capturing the relationships between variables.

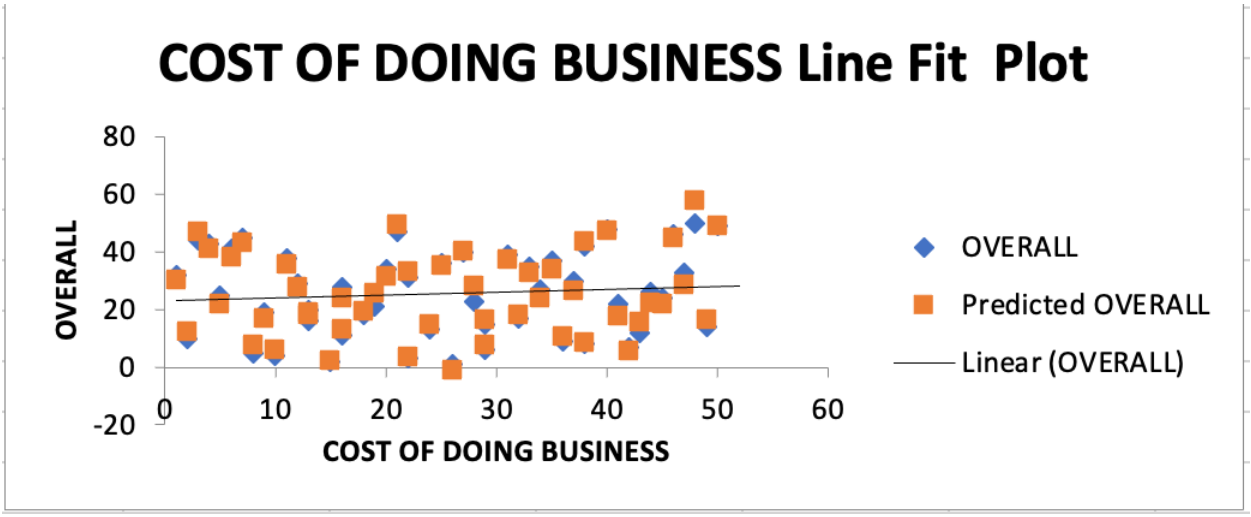
The adjusted R Square of 0.95706049 indicates that approximately 95.7% of the variability in the dependent variable is explained by the independent variables, further supporting the model's efficacy. The standard error of 3.02070286 suggests minimal errors in predicting the state ranks. (Bradburn, 2022) (Proud, 2021).

In summary, the results underscore the robustness of the regression model, showcasing its ability to effectively capture the interplay between the diverse factors influencing state rankings in CNBC's assessment.

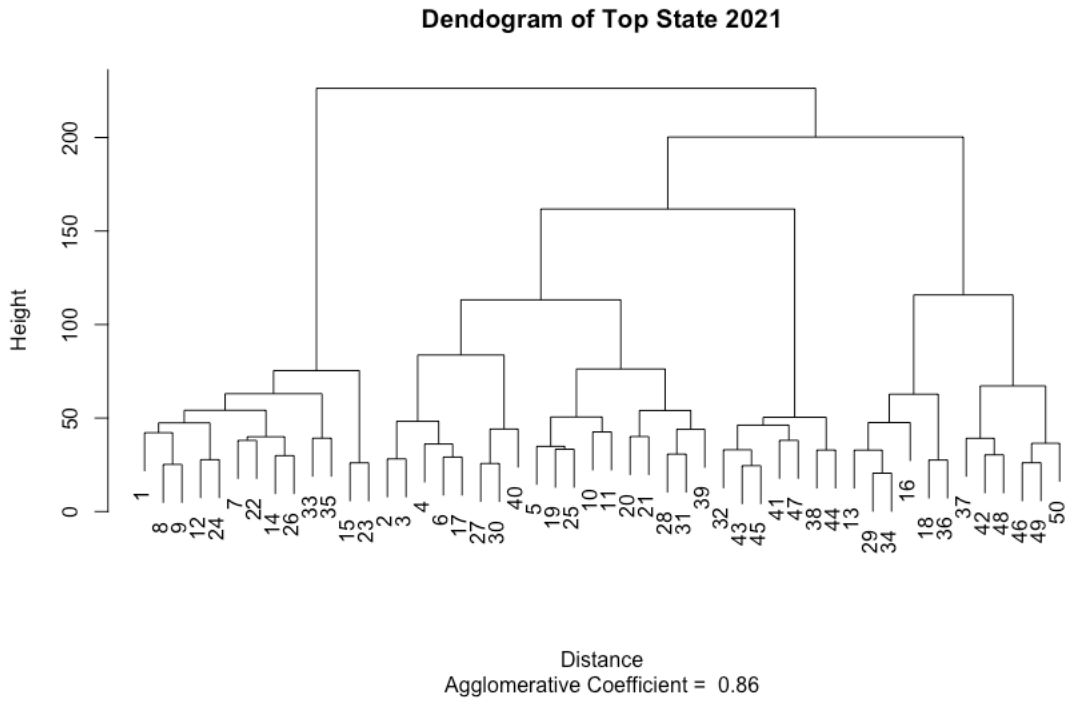
	Coefficients	*tandard Erro*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	-45.010458	3.0132326	-14.937598	1.003E-17	-51.10529615	-38.91562	-51.105296	-38.91562
COST OF DOING BUSINESS	0.23144	0.0642362	3.6029531	0.0008791	0.101510019	0.3613699	0.10151	0.3613699
INFRASTRUCTURE	0.3095914	0.0420287	7.3661983	6.769E-09	0.224580444	0.3946024	0.2245804	0.3946024
LIFE, HEALTH & INCLUSION	0.5084203	0.0545161	9.3260644	1.777E-11	0.398151132	0.6186894	0.3981511	0.6186894
WORKFORCE	0.2233746	0.051406	4.3453042	9.624E-05	0.119396222	0.3273531	0.1193962	0.3273531
ECONOMY	0.3063311	0.039899	7.6776569	2.562E-09	0.225627667	0.3870345	0.2256277	0.3870345
BUSINESS FRIENDLINESS	0.2419823	0.0425873	5.6820239	1.432E-06	0.155841282	0.3281234	0.1558413	0.3281234
ACCESS TO CAPITAL	0.3465085	0.0722792	4.7940259	2.394E-05	0.200309962	0.4927071	0.20031	0.4927071
TECHNOLOGY & INNOVATION	0.1907231	0.0529681	3.6007179	0.0008848	0.083585043	0.2978611	0.083585	0.2978611
EDUCATION	0.0705319	0.0488334	1.4443374	0.1566306	-0.028242977	0.1693068	-0.028243	0.1693068
COST OF LIVING	0.3532861	0.0608973	5.8013457	9.777E-07	0.230109728	0.4764624	0.2301097	0.4764624

The table represents an extension of the regression analysis, and for illustrative purposes, let us consider a specific variable, namely, the "Cost of Doing Business." The coefficient associated with this variable is 0.23144. In statistical terms, this coefficient signifies that if the ranking of the Cost of Doing Business were to transition from 1 to 2, the overall ranking would go

down by an average of 0.23, or equivalently, 23%. It is imperative to note that this analytical approach is applicable analogously to the remaining nine variables under consideration, namely, Infrastructure, Life Health and Inclusion, Workforce, Economy, Business Friendliness, Access to Capital, Technology and Innovation, Education, and Cost of Living.



According to the analysis of this scatter plot, the trendline appears to slope upwards, albeit with a relatively gentle incline. This suggests a positive correlation among the data points (Mario, 2017). To support this theoretical proposition, the application of hierarchical clustering analysis within R Studio yielded an agglomerative coefficient of 0.86. This value suggests a good natural clustering structure, thereby providing empirical support for the proposed hypothesis.



4. The Overall Ranking of Top States to Start a Business and Average Tax Ranking

4.1 Year of 2021

Top Ten Best States in 2021 based on CNBC's Evaluation Method

Upon juxtaposing two datasets, specifically the "Top State to Start a Business" rankings derived from CNBC's comprehensive assessment and the corresponding average tax rates, a discerning analysis unfolds. The ensuing table clarifies how the overall ranking, as per CNBC's evaluative metrics, could potentially influence the decision-making processes of discerning investors. For a more detailed exploration, the top 10 results based on CNBC's overall ranking for the best state to start a business in 2021 are shown below:

	OVERALL RANK	Best State to Start a Business	Tax Ranking
1	1	Virginia	6.00%
2	2	North Carolina	2.50%
3	3	Utah	4.95%
4	4	Texas	0.00%
5	5	Tennessee	6.50%
6	6	Georgia	5.75%
7	7	Minnesota	9.80%
8	8	Colorado	4.55%
9	9	Washington	0.00%
10	10	Ohio	0.00%

In the broader context of state evaluations for business initiation, Texas emerges as a noteworthy exemplar, maintaining its stature as one of the premier states for commencing business endeavors in 2021. A significant feature contributing to Texas's appeal is the absence of corporate taxes levied on businesses. Conversely, Ohio, while also devoid of corporate taxes, occupies a distinct position in the rankings, securing the 10th spot in the overall state-to-start-a-business hierarchy. The subtle evaluation of Ohio's business landscape reveals a noteworthy factor influencing its ranking: a relatively lower score in business friendliness, where it stands at the 47th position out of 50 states. This diminished ranking is attributed to the intricacies inherent in Ohio's legal and regulatory framework, manifesting in complexities related to trade, labor, and bureaucratic regulations (John, How we chose America's Top States for Business in 2021 , 2021). In the comprehensive evaluation of state performance metrics in 2021, Ohio amassed a cumulative total of 1423 points (Staff C. , 10. Ohio, 2021).

Washington does not impose corporate income tax, reflecting a pro-business stance. However, a deeper examination reveals that despite this tax advantage, Washington grapples with challenges. The state's Cost of Doing Business and Cost of Living indices are relatively high,

ranking 38th and 39th out of 50, respectively. Additionally, in the Business Friendliness category, which assesses the complexity of regulations, Washington secures the 39th position. This multifaceted evaluation collectively contributes to Washington's cumulative score of 1439 points in the year 2021, underscoring the intricacies involved in assessing a state's overall competitiveness (Staff C. , 9. Washington, 2021).

Minnesota emerges as a compelling case study within the top 10 states for business initiation, despite harboring one of the nation's highest corporate tax rates, earning it the 48th spot out of 50 states in this regard. While Minnesota's tax implications may seem counterintuitive, CNBC's comprehensive methodology reveals a nuanced perspective. With an overall tally of 1447 points, Minnesota grapples with a relatively high cost of living (ranking 30th) and intricate regulatory processes, reflected in its 28th position in the "Business Friendliness" category. Notably, Minnesota excels in "Life, Health, and Inclusion" (ranking 4th) and "Technology and Innovation" (ranking 4th) categories. Its education system also merits recognition, securing the 5th spot nationally in 2021. The intriguing paradox arises when considering that despite these accolades, skilled workers might opt to venture beyond the state's borders, possibly due to the higher cost of living in Minnesota. This nuanced analysis underscores the multifaceted nature of factors influencing Minnesota's appeal for business and workforce dynamics (Staff C. , 7. Minnesota, 2021).

Top Ten Worst States in 2021 based on CNBC's Evaluation Method

OVERALL RANK	Worst State to Start a Business	Tax Ranking
40	Nevada	0.00%
41	Kentucky	5.00%
42	Vermont	7.17%
43	Arkansas	3.87%
44	Louisiana	6.00%
45	Mississippi	4.00%
46	Rhode Island	7.00%
47	West Virginia	6.50%
48	Maine	7.17%
49	Hawaii	5.40%
50	Alaska	5.34%

Nevada, despite boasting a 0% corporate tax rate, surprisingly finds itself among the bottom 10 states to commence business operations. This unexpected placement is elucidated by a holistic examination of the state's overall score of 1190 points. Although Nevada enjoys the tax advantage of zero corporate tax, its deficiencies in crucial categories significantly impact its

standing. Notably, the state ranks at the bottom in the Technology and Innovation category, holding the 50th position. Additionally, subpar performances in education (49th rank) and Life, Health, and Inclusion (47th rank) further contribute to Nevada's descent into the bottom tier for business initiation. While Technology and Innovation hold a 7% weight in the overall ranking, the cumulative impact of education (6%) and Life, Health, and Inclusion (15%) pulls Nevada down among the least favorable states for business establishment. This underscores the multifaceted nature of evaluating states' business environments, where a singular tax advantage does not guarantee overall competitiveness (Staff C. , 40. Nevada, 2021).

Alaska, characterized by its frigid climate, emerges as an unsurprising contender for the least favorable state to initiate business endeavors. With a total score of 958 points out of 2500, Alaska grapples with one of the lowest overall category rankings. Nevertheless, it is noteworthy that Alaska manages to secure a comparatively better position in certain categories. The state's Life, Health, and Inclusion category secures the 19th rank, while the Business Friendliness category attains the 16th position. These higher rankings in specific domains, especially when contrasted with Nevada's performance, add nuance to Alaska's overall standing in the intricate landscape of states conducive to business initiation (Staff C. , 50. Alaska, 2021).

Amidst the cohort of states ranking within the bottom decile for business initiation, Maine distinguishes itself by boasting the highest corporate tax rate, a notable 7.17%. This financial imposition stands as a defining characteristic, setting Maine apart within the context of less favorable states for entrepreneurial endeavors.

4.2 Year of 2022

Top Ten Best States in 2022 based on CNBC's Evaluation Method

2022		
OVERALL RANK	Best State to Start a Business	Tax Rating
1	North Carolina	2.50%
2	Washington	0.00%
3	Virginia	6.00%
4	Colorado	4.55%
5	Texas	0.00%
6	Tennessee	6.50%
7	Nebraska	6.54%
8	Utah	4.85%
9	Minnesota	9.80%
10	Georgia	5.75%

In the year 2022, Washington witnessed a remarkable ascent, surging from the 9th position in 2021 to a commendable 2nd rank. This meteoric rise positions Washington as a prime candidate for initiating business endeavors, earning it the prestigious status of the best state to commence a

business in 2022. Notably, it secures the 2nd rank in CNBC's comprehensive evaluation of the overall top states for business initiation, coupled with the advantage of imposing a 0% corporate tax rate. Washington's commendable performance is reflected in its accumulated total of 1558 points. Despite these promising prospects, potential investors must navigate the challenge posed by Washington's relatively high cost of living, holding the 38th position out of 50 states in the overarching ranking (Staff C. , 9. Washington, 2021). To address this challenge, investors might choose to live outside of Washington while establishing their business operations in Washington. This strategic approach serves as an insightful consideration for investors contemplating the dynamic landscape of business initiation in Washington.

Revisiting the insights provided in the 2021 article concerning North Carolina, the state has exhibited a noteworthy resurgence in 2022. Building upon its commendable recovery trajectory observed in 2021, following the upheaval induced by the 2020 pandemic, North Carolina has emerged as the preeminent leader, securing the topmost position in the business initiation chart. Bolstering this achievement is the state's steadfast commitment to maintaining a consistent corporate tax rate of 2.5%, a policy in effect since 2018. This strategic fiscal approach has propelled North Carolina to a cumulative total of 1580 points in the evaluation. However, a critical appraisal reveals areas for potential enhancement, particularly in the Life, Health, and Inclusion category, where the state is positioned at number 28. Furthermore, North Carolina grapples with relatively high rankings in the domains of cost of living (22nd position) and business friendliness. These nuanced considerations underscore the multifaceted landscape of North Carolina's business environment, emphasizing the need for continued improvement in specific dimensions for sustained competitiveness (Staff C. , 1. North Carolina, 2022).

In conclusion, the discerning investor seeking an optimal environment for business initiation should meticulously evaluate the distinctive advantages offered by North Carolina and Washington. For entities prioritizing fiscal frugality and aiming to curtail corporate tax expenses, the judicious selection of Washington or Texas as the primary operational headquarters may prove best. On the contrary, for companies in pursuit of a comprehensive and well-rounded business environment, Washington emerges as an auspicious investment destination, where multifaceted savings and advantages across various dimensions within the state present a compelling proposition.

Top Ten Worst States in 2022 based on CNBC's Evaluation Method

OVERALL RANK	Worst State to Start a Business	Tax Rating
39	Nevada	0.00%
39	Connecticut	7.50%
41	Arkansas	3.38%
42	New Jersey	8.63%
43	Maine	8.40%
44	West Virginia	6.50%
45	Rhode Island	7.00%
46	New Mexico	5.35%
47	Hawaii	5.40%
48	Louisiana	5.50%
49	Alaska	5.34%
50	Mississippi	4.50%

In the year 2021, New Jersey held a moderately favorable position, securing the 26th rank. However, in the subsequent year of 2022, New Jersey experienced a significant decline, plummeting to the 42nd rank according to CNBC's evaluation. Notably, among the states positioned in the bottom decile by CNBC, New Jersey stands out with the highest corporate tax rate. Accumulating a total of 1179 points, New Jersey faces considerable challenges in economic stability, earning the dubious distinction of being ranked 50th in the nation in 2022. The state's credit ratings and overall financial outlook exhibit marked instability and uncertainty.

Moreover, New Jersey's Business Friendliness category languishes at the 47th position, indicative of intricate regulations governing trade and employment, creating hurdles for new investors. The Cost of Doing Business category fares little better, securing the 43rd rank, implying relatively high operational costs. Additionally, the Cost-of-Living category is positioned at number 40, further accentuating the state's relatively elevated living expenses (Staff C. , 42. New Jersey , 2022).

In summation, the comprehensive analysis presented in this chart underscores a discouraging outlook for prospective investors considering the initiation or expansion of a new company in New Jersey. The amalgamation of high corporate tax rates, economic instability, intricate business regulations, and elevated operational and living costs positions New Jersey unfavorably among the choices available to new investors. These factors collectively suggest that the state may not be optimal for those seeking a conducive and welcoming environment for business endeavors.

The concurrent rankings of Nevada and Connecticut at number 39 underscore a similarity in their overall performance, as both states accrued an equivalent total of 1221 points (Staff C. , 39. Connecticut, 2022) (Staff C. , 39. Nevada, 2022). Despite this parity, Nevada persists with persisting challenges, maintaining its status with the weakest education system, a stagnant technology and innovation sector (ranked 50 and 47, respectively). Conversely, Connecticut grapples with economic adversity, positioned at number 47, compounded by high costs of doing business and a relatively elevated cost of living, ranked at 47 and 43, respectively. In summative analysis, the collective data suggests that both Nevada and Connecticut may not be optimal choices or entrepreneurial ventures due to high costs and suboptimal developmental prospects.

4.3 The year of 2023

Top Ten Best States in 2022 based on CNBC's Evaluation Method

2023		
OVERALL RANK	Best State to Start a Business	Tax Rating
1	North Carolina	2.50%
2	Virginia	6.00%
3	Tennessee	6.50%
4	Georgia	5.75%
5	Minnesota	9.80%
6	Texas	0.00%
7	Washington	0.00%
8	Florida	5.50%
9	Utah	4.85%
10	Michigan	6.00%

Minnesota emerges as a notable case study within the evaluated states. Despite maintaining a consistent corporate tax rate of 9.8%, it exhibited an ascent in rank, progressing from number 9 to number 5. Over the course of the year, Minnesota experienced an increase in total points, advancing from 1466 to 1483. This subtle development raises questions about the correlation between tax policies and overall business environment, warranting further investigation into the state's specific economic dynamics and the interplay of factors contributing to its improved standing.

MINNESOTA SCORE AND RANKING

CATEGORY	2022 SCORE	2022 RANK
WORKFORCE	226	21
INFRASTRUCTURE	244	4
COST OF DOING BUSINESS	125	41
ECONOMY	200	16
LIFE & HEALTH	245	5
TECHNOLOGY & INNOVATION	179	6
BUSINESS FRIENDLINESS	88	32
EDUCATION	104	15
ACCESS TO CAPITAL	30	13
COST OF LIVING	25	26
OVERALL	1466	9

MINNESOTA SCORE AND RANKING

CATEGORY	2023 SCORE	2023 RANK
WORKFORCE	228	17
INFRASTRUCTURE	254	3
ECONOMY	193	24
LIFE, HEALTH & INCLUSION	269	4
COST OF DOING BUSINESS	103	39
TECHNOLOGY & INNOVATION	212	4
BUSINESS FRIENDLINESS	88	30
EDUCATION	75	13
ACCESS TO CAPITAL	28	20
COST OF LIVING	33	17
OVERALL	1483	5

The depicted chart clarifies a discernible trend wherein Minnesota's overall variable ranking exhibited improvement over the course of the year. This trajectory suggests a gradual and consistent improvisation in the state's business environment. This positive indication may serve as an inviting prospect for prospective investors considering business development in Minnesota, not only presently but also in the foreseeable future. It is imperative, however, to underscore that North Carolina retains a more robust standing and superior overall ranking, emphasizing its stronger positioning in comparison to Minnesota.

Top Ten Worst States in 2023 based on CNBC's Evaluation Method

2023		
OVERALL RANK	Best State to Start a Business	Tax Rating
40	New Hampshire	7.50%
41	Oklahoma	4.00%
42	Alabama	6.50%
43	New Mexico	5.35%
44	Arkansas	3.26%
45	Rhode Island	7.00%
46	West Virginia	6.50%
47	Hawaii	5.40%
48	Mississippi	4.50%
49	Louisiana	5.50%
50	Alaska	5.34%

New Hampshire, in the year 2022, secured the 35th position, yet experienced a notable descent to the 40th rank in 2023. The total points gathered mirrored this decline, diminishing from 1246 in 2022 to 1197 in 2023. This descending regression can be attributed to a confluence of factors, possibly intensified by the pervasive impact of the pandemic. The escalating prices in the region have engendered a rise in both the Cost of Living and the Cost of Doing Business, while simultaneously witnessing a reduction in the availability of skilled workers. Noteworthy is the reduction in corporate tax rating; however, this improvement did not translate into an overall enhancement in New Hampshire's variable rankings, underscoring the multifaceted challenges faced by the state (Staff C. , 40. New Hampshire, 2023) (Staff C. , 35. CNBC, 2022).

Arkansas witnessed a decline in its ranking from 41st place in 2022 to 44th in 2023. Notably, the corporate tax rates in the state experienced a marginal decrease, contracting from 3.38% in 2022 to 3.26%. While this rate is relatively low in comparison to counterparts such as New Hampshire or Rhode Island, Arkansas amassed a total of 1126 points. The discerning factor contributing to this downturn is evident in the state's Technology and Innovation category, securing the 49th rank – one of the lowest in the nation. Furthermore, the Business Friendliness category, positioned at number 44, implies that the landscape of labor, trades, and regulations in Arkansas tends towards complexity. This nuanced analysis underscores the intricacies influencing the state's overall business environment (Staff C. , 44. Arkansas, 2023).

In an overarching evaluation, Arkansas emerges as a less favorable option for initiating business ventures, particularly in the realm of technology companies. The state's diminished resources in technology and innovation, underscored by a ranking of 46, coupled with infrastructure challenges, contribute to its less-than-optimal standing in the broader spectrum of business-friendly environments. This nuanced examination sheds light on Arkansas's potential

limitations and serves as a valuable consideration for investors contemplating the establishment of technology-oriented enterprises within the state.

5. Forecasting Analysis

5.1 Virginia

According to CNBC's 2021 ranking, Virginia was deemed the top state for commencing business ventures. However, in 2022, Virginia's ranking dropped. By examining the data from 2021 to 2023, we can observe how Virginia's points decreased over time. Despite the limitations of this analysis, it provides valuable insights for economists and investors regarding the potential trajectory of states' business environments. It's important to note that the numbers presented represent points accumulated by the state rather than rankings.

	Virginia				
CATEGORY	2021	2022	2023	2024	2025
ACCESS TO CAPITAL	140	41	43	-22	-71
BUSINESS FRIENDLINESS	137	145	135	137	136
COST OF DOING BUSINESS	220	193	120	78	28
COST OF LIVING	29	21	28	25	25
ECONOMY	158	184	235	269	308
EDUCATION	119	129	94	89	77
INFRASTRUCTURE	191	235	231	259	279
LIFE, HEALTH & INCLUSION	245	204	217	194	180
TECHNOLOGY & INNOVATION	104	146	172	209	243
WORKFORCE	244	255	252	258	262
TOTAL	1587	1553	1527	1496	1466

Utilizing sophisticated analytical tools like Microsoft Excel's forecasting analysis, we can project Virginia's performance for the years 2024 and 2025. By employing the formula "=forecast(x, y-axis range, x-axis range)", where "x" represents the year being predicted (2024 & 2025), the "y-axis range" encompasses known variables such as access to capital, and the "x-axis range" denotes the previous years' data (2021-2023).

However, it's important to acknowledge the limitations inherent in forecast analyses. For instance, in the Access to Capital category, a negative point value of -22 was observed, which contradicts the minimum point range of 0. This discrepancy underscores a constraint of forecast analysis.

Nevertheless, based on the data from 2021 to 2023, the forecast suggests a projected increase of 2 points in the Business Friendliness category. However, the overall trend indicates a

reduction in total points for Virginia, implying a potential decline in its overall state ranking in the upcoming years.

5.2 North Carolina

Another state often lauded for its economic resilience is North Carolina, renowned for its rapid recovery following the 2020 pandemic. Presented below are the forecasted points for North Carolina in the years 2024 and 2025. It's important to note that the numbers presented represent points accumulated by the state rather than rankings.

North Carolina					
CATEGORY	2021	2022	2023	2024	2025
ACCESS TO CAPITAL	141	106	39	-7	-58
BUSINESS FRIENDLINESS	141	191	127	139	132
COST OF DOING BUSINESS	242	253	179	162	130
COST OF LIVING	42	29	24	14	5
ECONOMY	187	162	310	343	404
EDUCATION	90	45	83	66	62
INFRASTRUCTURE	200	214	224	237	249
LIFE, HEALTH & INCLUSION	145	192	149	166	168
TECHNOLOGY & INNOVATION	119	105	202	225	267
WORKFORCE	239	283	291	323	349
OVERALL	1546	1580	1628	1667	1708

Over the span of 2021 to 2023, North Carolina has shown consistent improvement, albeit with some notable reductions in points for specific variables such as the cost of living. This suggests a forecasted trend of increasing living costs in the state going forward. However, the forecast analysis predicts an overall rise in North Carolina's total points to 1708 by 2025. Consequently, North Carolina is expected to either ascend in the rankings of top states for starting a business or maintain its current number one position.

6. New York City

New York City, a city of worldwide interest. New York City, a city with a population of 7.6 million people. A city with an average household income of $113,315. New York City is the largest city in the United States since 1790, which is lead after by Los Angeles and Chicago (New York City, New York Population 2024). New York City has historically held a preeminent global status, widely acknowledged as the epitome of a cosmopolitan hub. This esteemed recognition, however, is concomitant with a commensurate elevation in pricing, rendering the cost of living in the metropolis notably higher than that observed in comparable locales.

Upon closer examination through rigorous computational analysis, to a surprise, it is discerned that New York City, in the year 2018, attained the 27th position in the ranking based on average tax rates. Subsequently, in 2019, Alabama supplanted its position, only for New York City to reclaim the 27th position from 2020 to 2021. Impressively, New York City maintained a constancy in its corporate tax rate at 6.50% until 2021. However, a departure from this equilibrium occurred in 2022, as the average corporate tax rate experienced an increment to 6.88%, leading to a descent in the rankings to the 32nd position. Counterintuitively, the average tax rate of New York City does not align with the anticipated levels that corporations might expect, given the city's global reputation and standing. This divergence prompts a reevaluation of the prevailing assumptions surrounding the perceived fiscal landscape of New York City within the broader global context.

2018			2019		
27	**New York**	**6.50%**	27	Alabama	6.50%
28	Tennessee	6.50%	**28**	**New York**	**6.50%**

2020			2021		
27	**New York**	**6.50%**	**27**	**New York**	**6.50%**
28	Tennessee	6.50%	28	Tennessee	6.50%

2022			2023		
27	Alabama	6.50%	27	Nebraska	6.42%
28	Tennessee	6.50%	28	Alabama	6.50%
29	West Virginia	6.50%	29	Tennessee	6.50%
30	Nebraska	6.54%	30	West Virginia	6.50%
31	Montana	6.75%	31	Montana	6.75%
32	**New York**	**6.88%**	**32**	**New York**	**6.88%**
33	Rhode Island	7.00%	33	Iowa	6.95%

In recent years, New York has witnessed notable fluctuations in its rankings, initially starting at number 22 in 2021, indicating it was an above-average state for business commencement. However, there was a decline to number 36 in 2022, followed by a remarkable resurgence to number 20 in 2023. These shifts demonstrate a significant improvement within a relatively short timeframe, occurring just two years after the onset of the 2020 pandemic.

By leveraging Excel's forecasting analysis, economists and investors can anticipate New York's performance for the years 2024 and 2025.

	New York				
CATEGORY	2021	2022	2023	2024	2025
ACCESS TO CAPITAL	164	43	39	-43	-106
BUSINESS FRIENDLINESS	25	63	54	76	91
COST OF DOING BUSINESS	147	123	101	78	55
COST OF LIVING	3	2	9	11	14
ECONOMY	159	143	241	263	304
EDUCATION	102	121	75	72	59
INFRASTRUCTURE	217	189	222	214	217
LIFE, HEALTH & INCLUSION	218	176	228	217	222
TECHNOLOGY & INNOVATION	120	196	208	263	307
WORKFORCE	191	176	174	163	155
OVERALL	1346	1232	1351	1315	1317

According to the findings, New York's performance is expected to remain relatively stable. While no significant improvements or deteriorations are anticipated, there is a projected decline in New York's rankings for 2024. This decline may be attributed to delayed ripple effects from the pandemic, particularly impacting real estate and the cost of living, which could gradually improve by 2025. Consequently, residents of New York need not be overly concerned about dramatic changes in the near future. This also encourages investors to consider expanding their businesses in New York, despite the higher corporate tax rates, due to its stability.

Limitations of Study and Future Work

The study encountered limitations regarding the methodology employed. Specifically, while ten variables were considered in ranking the top states for commencing business endeavors, there was a lack of clarity regarding the process for assigning points to each variable. The overall points used for ranking were obtained from CNBC's website; however, the specific method by which these points were derived was not specified. For instance, in the case of Alaska receiving 216 out of 400 points for the cost of doing business in 2021, CNBC did not provide an explanation for how these 216 points were determined. This lack of transparency in the accumulation of points from the variables resulted in a vague measurement, thus limiting the comprehensiveness of the study.

The forecasting analysis conducted in this study was incomplete, as it only encompassed three states: Virginia, North Carolina, and New York. Consequently, the study failed to provide a comprehensive overview of the overall point results for all states, rendering it incomplete in its ability to rank states for the forecasted years of 2024 and 2025.

Furthermore, forecasting analysis inherently possesses limitations. As a predictive tool, its projections are not infallible and may deviate from actual outcomes. Moreover, unforeseen events such as the 2020 Covid-19 pandemic cannot be accurately predicted or factored into the analysis. Additionally, the presence of negative points in the forecast analysis is inconsistent with the point system's minimum value of 0, highlighting a discrepancy between forecasted values and actual constraints.

This study encompasses a wealth of data and information, enabling the generation of extensive economic predictions for both local and national contexts. It serves as a foundation for further investigation into the intricate mechanisms underlying the accumulation of points for each variable, such as the case of Alaska achieving 216 points out of a possible 400. This prompts several pertinent questions:

1. What are the criteria and prerequisites that facilitated Alaska’s attainment of 216 points?
2. How can Alaska enhance its performance to accrue higher point totals?
3. Despite Alaska’s challenging weather conditions, it ranked number 48 in the Cost of Doing Business category in 2021. What factors contributed to this ranking and how did other states fare comparatively?
4. What are the key determinants influencing these outcomes?

Exploring these inquiries in detail can provide valuable insights into the underlying dynamics shaping the rankings and performance of states in various economic categories.

The forecasting analysis is limited to the years 2021-2023 as a precautionary measure, considering the possibility of future pandemics like the one experienced in 2020. To further

explore this study, economists and researchers can extend the analysis by incorporating data from additional preceding years and applying the same point system. This expanded timeframe holds the potential to enhance the precision of forecasting analysis, offering investors and economists improved insights for predicting future trends and developments.

Conclusions

In conclusion, corporate tax rates represent just one aspect among several to contemplate when initiating a business venture. Take New York, for instance. Despite its perceived high tax burden, investors and entrepreneurs continue to regard it favorably due to its robust economic stability. However, the high costs associated with operating in New York may present challenges, particularly for first-time business owners. This underscores the significance of CNBC's methodology, which evaluates various variables crucial for establishing a resilient business environment. These evaluations have gained added importance following the disruptive economic repercussions of the 2020 Covid-19 pandemic, which precipitated abrupt shutdowns worldwide.

In essence, launching a business necessitates careful consideration of numerous factors. The data presented in the addendum charts assumes pivotal significance in guiding entrepreneurs' decisions on where to invest. Additionally, investors must weigh whether to initiate a new business or expand an existing one, as overall state ratings can significantly impact future business success. The occurrence of a pandemic represents an outlier event capable of halting an entire country's economic activity. However, establishing a company in a stable economic environment can serve as a protective measure against bankruptcy, provided it is managed effectively.

Although Excel's forecasting analysis is a valuable tool, it does not anticipate fluctuations in the economy. Consequently, investors should exercise caution and implement measures to mitigate risks when initiating new business ventures or expanding operations, particularly in light of unpredictable events.

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Addendum

Complete table of overall top state to start a business ranking and average tax rating for 2021.

2021		
OVERALL RANK	Best State to Start a Business	Tax Ranking
1	Virginia	6.00%
2	North Carolina	2.50%
3	Utah	4.95%
4	Texas	0.00%
5	Tennessee	6.50%
6	Georgia	5.75%
7	Minnesota	9.80%
8	Colorado	4.55%
9	Washington	0.00%
10	Ohio	0.00%
11	Michigan	6.00%
12	Maryland	8.25%
13	Nebraska	6.70%
14	Massachusetts	8.00%
15	Illinois	9.50%
16	Idaho	6.93%
17	Florida	4.46%
18	North Dakota	3.09%
19	Indiana	5.25%
20	Iowa	8.10%
21	Wisconsin	7.90%
22	New York	6.50%
23	Pennsylvania	9.99%
24	Connecticut	7.50%
25	Missouri	4.00%

2021		
OVERALL RANK	Best State to Start a Business	Tax Ranking
26	New Jersey	8.63%
27	Delaware	8.70%
28	Kansas	5.50%
29	South Dakota	-
30	Arizona	4.90%
31	Alabama	6.50%
32	Oklahoma	6.00%
33	California	8.84%
34	Montana	6.75%
35	Oregon	6.60%
36	Wyoming	-
37	New Hampshire	7.70%
38	New Mexico	5.35%
39	South Carolina	5.00%
40	Nevada	0.00%
41	Kentucky	5.00%
42	Vermont	7.17%
43	Arkansas	3.87%
44	Louisiana	6.00%
45	Mississippi	4.00%
46	Rhode Island	7.00%
47	West Virginia	6.50%
48	Maine	7.17%
49	Hawaii	5.40%
50	Alaska	5.34%

Complete table of overall top state to start a business ranking and average tax rating for 2022.

2022		
OVERALL RANK	Best State to Start a Business	Tax Rating
1	North Carolina	2.50%
2	Washington	0.00%
3	Virginia	6.00%
4	Colorado	4.55%
5	Texas	0.00%
6	Tennessee	6.50%
7	Nebraska	6.54%
8	Utah	4.85%
9	Minnesota	9.80%
10	Georgia	5.75%
11	Florida	5.50%
12	Iowa	8.10%
13	North Dakota	3.09%
14	Indiana	4.90%
15	Ohio	0.00%
16	Michigan	6.00%
17	Pennsylvania	9.99%
18	Oregon	7.10%
19	Illinois	9.50%
20	Idaho	6.00%
21	Kansas	5.50%
22	South Dakota	-
23	Wisconsin	7.90%
24	Massachusetts	8.00%
25	Missouri	4.00%

2022		
OVERALL RANK	Best State to Start a Business	Tax Ranking
26	Kentucky	5.00%
27	Maryland	8.25%
28	Delaware	8.70%
29	California	8.84%
30	Montana	6.75%
31	Vermont	7.17%
32	Wyoming	-
33	Alabama	6.50%
34	Arizona	4.90%
35	New Hampshire	7.60%
36	New York	6.88%
36	South Carolina	5.00%
38	Oklahoma	4.00%
39	Nevada	0.00%
39	Connecticut	7.50%
41	Arkansas	3.38%
42	New Jersey	8.63%
43	Maine	8.40%
44	West Virginia	6.50%
45	Rhode Island	7.00%
46	New Mexico	5.35%
47	Hawaii	5.40%
48	Louisiana	5.50%
49	Alaska	5.34%
50	Mississippi	4.50%

Complete table of overall top state to start a business ranking and average tax rating for 2023.

2023		
OVERALL RANK	Best State to Start a Business	Tax Rating
1	North Carolina	2.50%
2	Virginia	6.00%
3	Tennessee	6.50%
4	Georgia	5.75%
5	Minnesota	9.80%
6	Texas	0.00%
7	Washington	0.00%
8	Florida	5.50%
9	Utah	4.85%
10	Michigan	6.00%
11	Colorado	4.40%
12	Ohio	0.00%
13	Indiana	4.90%
14	Arizona	4.90%
15	Pennsylvania	8.99%
15	Massachusetts	8.00%
17	Illinois	9.50%
18	Delaware	8.70%
19	New Jersey	8.63%
20	New York	6.88%
21	Oregon	7.10%
22	Maryland	8.25%
23	Kansas	5.50%
23	Iowa	6.95%
25	California	8.84%

2023		
OVERALL RANK	Best State to Start a Business	Tax Ranking
26	Idaho	5.80%
27	South Carolina	5.00%
28	Nebraska	6.42%
29	Vermont	7.17%
30	Nevada	0.00%
31	Connecticut	7.50%
32	Missouri	4.00%
33	Wisconsin	7.90%
34	South Dakota	-
35	Montana	6.75%
36	North Dakota	3.09%
37	Wyoming	-
38	Kentucky	5.00%
39	Maine	7.17%
40	New Hampshire	7.50%
41	Oklahoma	4.00%
42	Alabama	6.50%
43	New Mexico	5.35%
44	Arkansas	3.26%
45	Rhode Island	7.00%
46	West Virginia	6.50%
47	Hawaii	5.40%
48	Mississippi	4.50%
49	Louisiana	5.50%
50	Alaska	5.34%

2018 Tax Brackets and Regulations

State	Category	Rates		Brackets	Average Rate
Alabama		6.50%	>	$0	6.50%
Alaska		0.00%	>	$0	5.34%
Alaska		2.00%	>	$25,000	5.34%
Alaska		3.00%	>	$49,000	5.34%
Alaska		4.00%	>	$74,000	5.34%
Alaska		5.00%	>	$99,000	5.34%
Alaska		6.00%	>	$124,000	5.34%
Alaska		7.00%	>	$148,000	5.34%
Alaska		8.00%	>	$173,000	5.34%
Alaska		9.00%	>	$198,000	5.34%
Alaska		9.40%	>	$222,000	5.34%
Arizona		4.90%	>	$0	4.90%
Arkansas	a	2.00%	>	$3,000	3.83%
Arkansas	a	3.00%	>	$6,000	3.83%
Arkansas	a	5.00%	>	$11,000	3.83%
Arkansas	a	6.00%	>	$25,000	3.83%
Arkansas	a	5.95%	>	$100,000	3.83%
Arkansas	a	1.00%	>	$0	3.83%
California		8.84%	>	$0	8.84%
Colorado		4.63%	>	$0	4.63%
Connecticut	b	8.25%	>	$0	8.25%
Delaware	c	8.70%	>	$0	8.70%
Florida		5.50%	>	$0	5.50%
Georgia		6.00%	>	$0	6.00%
Hawaii		4.40%	>	$0	5.40%
Hawaii		5.40%	>	$25,000	5.40%
Hawaii		6.40%	>	$100,000	5.40%
Idaho		7.40%	>	$0	7.40%
Illinois	d	9.50%	>	$0	9.50%
Indiana	e	6.00%	>	$0	6.00%
Iowa		6.00%	>	$0	9.00%
Iowa		8.00%	>	$25,000	9.00%
Iowa		10.00%	>	$100,000	9.00%
Iowa		12.00%	>	$250,000	9.00%

Kansas		4.00%	>	$0	5.50%
Kansas		7.00%	>	$50,000	5.50%
Kentucky		4.00%	>	$0	5.00%
Kentucky		5.00%	>	$50,000	5.00%
Kentucky		6.00%	>	$100,000	5.00%
Louisiana		4.00%	>	$0	6.00%
Louisiana		5.00%	>	$25,000	6.00%
Louisiana		6.00%	>	$50,000	6.00%
Louisiana		7.00%	>	$100,000	6.00%
Louisiana		8.00%	>	$200,000	6.00%
Maine		3.50%	>	$0	7.17%
Maine		7.93%	>	$25,000	7.17%
Maine		8.33%	>	$75,000	7.17%
Maine		8.93%	>	$250,000	7.17%
Maryland		8.25%	>	$0	8.25%
Massachusetts		8.00%	>	$0	8.00%
Michigan		6.00%	>	$0	6.00%
Minnesota		9.80%	>	$0	9.80%
Mississippi		0.00%	>	$0	3.00%
Mississippi		3.00%	>	$1,000	3.00%
Mississippi		4.00%	>	$5,000	3.00%
Mississippi		5.00%	>	$10,000	3.00%
Missouri		6.25%	>	$0	6.25%
Montana		6.75%	>	$0	6.75%
Nebraska		5.58%	>	$0	6.70%
Nebraska		7.81%	>	$100,000	6.70%
Nevada	c		(c)		0
New Hampshire		8.20%	>	$0	8.20%
New Jersey	f	9.00%	>	$100,000	9.00%
New Mexico	g	4.80%	>	$0	5.35%
New Mexico	g	5.90%	>	$500,000	5.35%
New York		6.50%	>	$0	6.50%
North Carolina		3.00%	>	$0	3.00%
North Dakota		1.41%	>	$0	3.09%
North Dakota		3.55%	>	$25,000	3.09%
North Dakota		4.31%	>	$50,000	3.09%
Ohio	c		(c)		0
Oklahoma		6.00%	>	$0	6.00%
Oregon		6.60%	>	$0	7.10%

Oregon		7.60%	>	$1,000,000	7.10%
Pennsylvania		9.99%	>	$0	9.99%
Rhode Island		7.00%	>	$0	7.00%
South Carolina		5.00%	>	$0	5.00%
South Dakota			None		n/a
Tennessee		6.50%	>	$0	6.50%
Texas	c		(c)		0
Utah		5.00%	>	$0	5.00%
Vermont		6.00%	>	$0	7.17%
Vermont		7.00%	>	$10,000	7.17%
Vermont		8.50%	>	$25,000	7.17%
Virginia	c	6.00%	>	$0	6.00%
Washington	c		(c)		0
Washington, D.C.		8.25%	>	$0	8.25%
West Virginia		6.50%	>	$0	6.50%
Wisconsin		7.90%	>	$0	7.90%
Wyoming			None		n/a

(a) 2018 -In Arkansas, corporations with net income over $100,000 pay 5.4 percent on the first $100,000 plus 6.5 percent on the excess over $100,000.
(b) 2018 - Rate includes a 10% surtax, which effectively increases the rate from 7.5% to 8.25%. Surtax is required by businesses with at least $100 million annual gross income
(c) 2018 - Nevada, Ohio, Texas, and Washington do not have a corporate income tax but do have a gross receipts tax with rates not strictly comparable to corporate income tax rates. See Table 18 for more information. Delaware and Virginia have gross receipts taxes in addition to corporate income taxes
(d) 2018 - Illinois' rate includes two separate corporate income taxes, one at a 7.0% rate and one at a 2.5% rate.
(e) 2018 - The tax rate in Indiana will decrease to 5.75% on July 1, 2018.
(f) 2018 - Corporations with entire net income greater than $100,000 pay 9% on all taxable income, companies with entire net income greater than $50,000 and less than or equal to $100,000 pay 7.5% on all taxable income, and companies with entire net income less than or equal to $50,000 pay 6.5% on all taxable income.

Note: In addition to regular income taxes, many states impose other taxes on corporations such as gross receipts taxes and franchise taxes. Some states also impose an alternative minimum tax and special rates on financial institutions.

2019 Tax Brackets and Regulations

State	Category	Rates		Brackets	Average Rate
Alabama		6.50%	>	$0	6.50%
Alaska		6.50%	>	$0	5.45%
Alaska		0.00%	>	$0	5.45%
Alaska		2.00%	>	$25,000	5.45%
Alaska		3.00%	>	$49,000	5.45%
Alaska		4.00%	>	$74,000	5.45%
Alaska		5.00%	>	$99,000	5.45%
Alaska		6.00%	>	$124,000	5.45%
Alaska		7.00%	>	$148,000	5.45%
Alaska		8.00%	>	$173,000	5.45%
Alaska		9.00%	>	$198,000	5.45%
Alaska		9.40%	>	$222,000	5.45%
Arizona		4.90%	>	$0	4.90%
Arkansas		1.00%	>	$0	3.92%
Arkansas		2.00%	>	$3,000	3.92%
Arkansas		3.00%	>	$6,000	3.92%
Arkansas		5.00%	>	$11,000	3.92%
Arkansas		6.00%	>	$25,000	3.92%
Arkansas		6.50%	>	$100,000	3.92%
California		8.84%	>	$0	8.84%
Colorado		4.63%	>	$0	4.63%
Connecticut		7.50%	>	$0	7.50%
D.C.		8.25%	>	$0	8.25%
Delaware	a	8.70%	>	$0	8.70%
Florida		5.50%	>	$0	5.50%
Georgia	b	5.75%	>	$0	5.75%
Hawaii		4.40%	>	$0	5.18%
Hawaii		5.40%	>	$25,000	5.18%
Hawaii		6.40%	>	$100,000	5.18%
Idaho		6.93%	>	$0	6.93%
Illonois	c	9.50%	>	$0	9.50%
Indiana	d	5.75%	>	$0	5.75%
Iowa	e	6%	>	$0	8%
Iowa	e	12%	>	$250,000	8%

Iowa	e	8%	>	$25,000	8%
Iowa	e	10%	>	$100,000	8%
Kansas		4%	>	$0	7%
Kansas		7%	>	$50,000	7%
Kentucky		5%	>	$0	5%
Louisiana		4%	>	$0	5%
Louisiana		5%	>	$25,000	5%
Louisiana		6%	>	$50,000	5%
Louisiana		7%	>	$100,000	5%
Louisiana		8%	>	$200,000	5%
Maine		3.50%	>	$0	6.94%
Maine		7.93%	>	$350,000	6.94%
Maine		8.33%	>	$1,050,000	6.94%
Maine		8.93%	>	$3,500,000	6.94%
Maryland		8.25%	>	$0	8.25%
Massachusetts		8%	>	$0	8%
Michigan		6%	>	$0	6%
Minnesota		9.80%	>	$0	9.80%
Mississippi	f	3%	>	$2,000	6%
Mississippi	f	4%	>	$5,000	6%
Mississippi	f	5%	>	$10,000	6%
Missouri	g	6.25%	>	$0	6.25%
Montana		6.75%	>	$0	6.75%
Nebraska		5.58%	>	$0	6.17%
Nebraska		7.81%	>	$100,000	6.17%
Nevada	a		(a)		0
New Hampshire		7.70%	>	$0	7.70%
New Jersey	h	6.50%	>	$0	7.68%
New Jersey	h	7.50%	>	$50,000	7.68%
New Jersey	h	9.00%	>	$100,000	7.68%
New Jersey	h	11.50%	>	$1,000,000	7.68%
New Mexico		4.80%	>	$0	8.15%
New Mexico		5.90%	>	$500,000	8.15%
New York		6.50%	>	$0	6.50%
North Carolina		2.50%	>	$0	2.50%
North Dakota		1.41%	>	$0	2.49%
North Dakota		3.55%	>	$25,000	2.49%
North Dakota		4.31%	>	$50,000	2.49%

Ohio	a		(a)		0
Oklahoma		6%	>	$0	6%
Oregon		6.60%	>	$0	6.30%
Oregon		7.60%	>	$1,000,000	6.30%
Pennsylvania		9.99%	>	$0	9.99%
Rhode Island		7%	>	$0	7%
South Carolina		5%	>	$0	5%
South Dakota			None		-
Tennessee		6.50%	>	$0	6.50%
Texas	a		(a)		0
Utah		4.95%	>	$0	4.95%
Vermont		6.00%	>	$0	5.48%
Vermont		7.00%	>	$10,000	5.48%
Vermont		8.50%	>	$25,000	8.50%
Virginia	a	6%	>	$0	6%
Washington	a		(a)		0
West Virginia		6.50%	>	$0	6.50%
Wisconsin		7.90%	>	$0	7.90%
Wyoming			None		-

(a) 2019 - Nevada, Ohio, Texas, and Washington do not have a corporate income tax but do have a gross receipts tax with rates not strictly comparable to corporate income tax rates. Delaware has gross receipts taxes in addition to corporate income taxes, as do several states like Pennsylvania, Virginia, and West Virginia, which permit gross receipts taxes at the local (but not state) level.
(b) 2019 - Georgia's corporate income tax rate will revert to 6% on January 1, 2026. The state could see a drop to 5.5% in 2019, pending legislative approval.
(c) 2019 - Illinois' rate includes two separate corporate income taxes, one at a 7% rate and one at a 2.5% rate.
(d) 2019 - Indiana's rate will change to 5.5% on July 1, 2019. The rate is scheduled to decrease to 4.9% by 2022.
(e) 2019 - Iowa's rate is scheduled to drop to 9.8 percent by 2021, subject to revenue availability.
(f) 2019 - Mississippi continues to phase out the 3 percent bracket by increasing the exemption by $1,000 a year. By the start of 2022, the 3 percent bracket will be fully eliminated.
(g) 2019 - Scheduled reform in 2020 will subject nearly all Missouri companies to a single sales factor appointment, permitting a rate reduction from 6.25% to 4%.
(h) 2019 - In New Jersey, the rates indicated apply to a corporation's entire net income rather than just income over the threshold. A temporary surcharge is in effect, bringing the rate to 11.5 percent for businesses with income over $1 million. Note: In addition to regular income taxes,

many states impose other taxes on corporations such as gross receipts taxes and franchise taxes. Some states also impose an alternative minimum tax and special rates on financial institutions.

2020 Tax Brackets and Regulations

State	Category	Rates		Brackets	Rate
Alabama		6.50%	>	$0	6.50%
Alaska		0.00%	>	$0	5.34%
Alaska		2.00%	>	$25,000	5.34%
Alaska		3.00%	>	$49,000	5.34%
Alaska		4.00%	>	$74,000	5.34%
Alaska		5.00%	>	$99,000	5.34%
Alaska		6.00%	>	$124,000	5.34%
Alaska		7.00%	>	$148,000	5.34%
Alaska		8.00%	>	$173,000	5.34%
Alaska		9.00%	>	$198,000	5.34%
Alaska		9.40%	>	$222,000	5.34%
Arizona		4.90%	>	$0	4.90%
Arkansas		1.00%	>	$0	3.92%
Arkansas		2.00%	>	$3,000	3.92%
Arkansas		3.00%	>	$6,000	3.92%
Arkansas		5.00%	>	$11,000	3.92%
Arkansas		6.00%	>	$25,000	3.92%
Arkansas		6.50%	>	$100,000	3.92%
California		8.84%	>	$0	8.84%
Colorado		4.63%	>	$0	4.63%
Connecticut		7.50%	>	$0	7.50%
Delaware	a	8.70%	>	$0	8.70%
Florida		4.46%	>	$0	4.46%
Georgia	c	5.75%	>	$0	5.75%
Hawaii		4.40%	>	$0	5.40%
Hawaii		5.40%	>	$25,000	5.40%
Hawaii		6.40%	>	$100,000	5.40%
Idaho		6.93%	>	$0	6.93%
Illinois	d	9.50%	>	$0	9.50%
Indiana	e	5.50%	>	$0	5.50%
Iowa		6%	>	$0	9%
Iowa		8%	>	$25,000	9%

Iowa		10%	>	$100,000	9%
Iowa		12%	>	$250,000	9%
Kansas		4%	>	$0	6%
Kansas		7%	>	$50,000	6%
Kentucky		5%	>	$0	5%
Louisiana		4%	>	$0	6%
Louisiana		5%	>	$25,000	6%
Louisiana		6%	>	$50,000	6%
Louisiana		7%	>	$100,000	6%
Louisiana		8%	>	$200,000	6%
Maine		3.50%	>	$0	7.17%
Maine		7.93%	>	$350,000	7.17%
Maine		8.33%	>	$1,050,000	7.17%
Maine		8.93%	>	$3,500,000	7.17%
Maryland		8.25%	>	$0	8.25%
Massachusetts		8%	>	$0	8%
Michigan		6%	>	$0	6%
Minnesota		9.80%	>	$0	9.80%
Mississippi	f	3%	>	$3,000	4%
Mississippi	f	4%	>	$5,000	4%
Mississippi	f	5%	>	$10,000	4%
Missouri		4.00%	>	$0	4.00%
Montana		6.75%	>	$0	6.75%
Nebraska		5.58%	>	$0	6.70%
Nebraska		7.81%	>	$100,000	6.70%
Nevada	a		(a)		0
New Hampshire	g	7.70%	>	$0	7.70%
New Jersey	h	6.50%	>	$0	8.38%
New Jersey	h	7.50%	>	$50,000	8.38%
New Jersey	h	9.00%	>	$100,000	8.38%
New Jersey	h	10.50%	>	$1,000,000	8.38%
New Mexico		4.80%	>	$0	5.35%
New Mexico		5.90%	>	$500,000	5.35%
New York		6.50%	>	$0	6.50%
North Carolina		2.50%	>	$0	2.50%
North Dakota		1.41%	>	$0	3.09%
North Dakota		3.55%	>	$25,000	3.09%
North Dakota		4.31%	>	$50,000	3.09%
Ohio	a		(a)		0

Oklahoma		6%	>	$0	6%
Oregon	a	6.60%	>	$0	7.10%
Oregon	a	7.60%	>	$1,000,000	7.10%
Pennsylvania		9.99%	>	$0	9.99%
Rhode Island		7%	>	$0	7%
South Carolina		5%	>	$0	5%
South Dakota			None		-
Tennessee		6.50%	>	$0	6.50%
Texas	a		(a)		0
Utah		4.95%	>	$0	4.95%
Vermont		6.00%	>	$0	7.17%
Vermont		7.00%	>	$10,000	7.17%
Vermont		8.50%	>	$25,000	7.17%
Virginia	a	6%	>	$0	6%
Washington	a		(a)		0
Washington, D.C.		8.25%	>	$0	8.25%
West Virginia		6.50%	>	$0	6.50%
Wisconsin		7.90%	>	$0	7.90%
Wyoming			None		-

(a) 2020 - Nevada, Ohio, Texas, and Washington do not have a corporate income tax but do have a gross receipts tax with rates not strictly comparable to corporate income tax rates. See Table 18 for more information. Delaware and Oregon have gross receipts taxes in addition to corporate income taxes, as do several states like Pennsylvania, Virginia, and West Virginia, which permit gross receipts taxes at the local (but not state) level.
(b) 2020 - Florida's corporate income tax rate will return to 5.5% for tax years beginning on or after Jan. 1, 2022.
(c) 2020 - Georgia's corporate income tax rate will revert to 6% on January 1, 2026. The state could see a drop to 5.5% in 2020, pending legislative approval.
(d) 2020 - Illinois' rate includes two separate corporate income taxes, one at a 7% rate and one at a 2.5% rate.
(e) 2020 - Indiana's rate will change to 5.25% on July 1, 2020. The rate is scheduled to decrease to 4.9% by 2022.
(f) 2020 - Mississippi continues to phase out the 3 percent bracket by increasing the exemption by $1,000 a year. This year, the exemption is $3,000. By the start of 2022, the 3 percent bracket will be fully eliminated.
(g) 2020 - New Hampshire's rate is 7.9% for tax periods ending before Dec. 31, 2019.

(h) 2020 - In New Jersey, the rates indicated apply to a corporation’s entire net income rather than just income over the threshold. A temporary surcharge is in effect, bringing the rate to 10.5 percent for businesses with income over $1 million.

2021 Tax Brackets and Regulations

State	Category	Rates		Brackets	Rate
Alabama		6.5%	>	$0	6.5%
Alaska		0%	>	$0	5%
Alaska		2%	>	$25,000	5%
Alaska		3%	>	$49,000	5%
Alaska		4%	>	$74,000	5%
Alaska		5%	>	$99,000	5%
Alaska		6%	>	$124,000	5%
Alaska		7%	>	$148,000	5%
Alaska		8%	>	$173,000	5%
Alaska		9%	>	$198,000	5%
Alaska		9.4%	>	$222,000	5%
Arizona		4.9%	>	$0	4.9%
Arkansas		1.0%	>	$0	3.9%
Arkansas		2.0%	>	$3,000	3.9%
Arkansas		3.0%	>	$6,000	3.9%
Arkansas		5.0%	>	$11,000	3.9%
Arkansas		6.0%	>	$25,000	3.9%
Arkansas		6.2%	>	$100,000	3.9%
California		8.84%	>	$0	8.84%
Colorado		4.55%	>	$0	4.55%
Connecticut		7.50%	>	$0	7.50%
Delaware	a	8.7%	>	$0	8.7%
Florida	b	4.458%	>	$0	4.458%
Georgia	c	5.75%	>	$0	5.75%
Hawaii		4.4%	>	$0	5.4%
Hawaii		5.4%	>	$25,000	5.4%
Hawaii		6.4%	>	$100,000	5.4%
Idaho		6.925%	>	$0	6.925%
Illonois	d	9.50%	>	$0	9.50%
Indiana	e	5.25%	>	$0	5.25%
Iowa		5.5%	>	$0	8.1%
Iowa		9.0%	>	$100,000	8.1%
Iowa		9.8%	>	$250,000	8.1%
Kansas		4%	>	$0	6%
Kansas		7%	>	$50,000	6%

Kentucky		5%	>	$0	5%
Louisiana		4%	>	$0	6%
Louisiana		5%	>	$25,000	6%
Louisiana		6%	>	$50,000	6%
Louisiana		7%	>	$100,000	6%
Louisiana		8%	>	$200,000	6%
Maine		3.50%	>	$0	7.17%
Maine		7.93%	>	$350,000	7.17%
Maine		8.33%	>	$1,050,000	7.17%
Maine		8.93%	>	$3,500,000	7.17%
Maryland		8.25%	>	$0	8.25%
Massachusetts		8.00%	>	$0	8.00%
Michigan		6.00%	>	$0	6.00%
Minnesota		9.8%	>	$0	9.8%
Mississippi	f	3%	>	$4,000	4%
Mississippi	f	4%	>	$5,000	4%
Mississippi	f	5%	>	$10,000	4%
Missouri		4.00%	>	$0	4.00%
Montana		6.75%	>	$0	6.75%
Nebraska		5.58%	>	$0	6.70%
Nebraska		7.81%	>	$100,000	6.70%
Nevada	a		(a)		0
New Hamsphire		7.7%	>	$0	7.7%
New Jersey	g	6.5%	>	$0	8.6%
New Jersey	g	7.5%	>	$50,000	8.6%
New Jersey	g	9.0%	>	$100,000	8.6%
New Jersey	g	11.5%	>	$1,000,000	8.6%
New Mexico		4.8%	>	$0	5.4%
New Mexico		5.9%	>	$500,000	5.4%
New York		6.5%	>	$0	6.5%
North Carolina		2.5%	>	$0	2.5%
North Dakota		1.41%	>	$0	3.09%
North Dakota		3.55%	>	$25,000	3.09%
North Dakota		4.31%	>	$50,000	3.09%
Ohio	a		(a)		0.00%
Oklahoma		6%	>	$0	6%
Oregon	a	6.6%	>	$0	6.6%
Oregon	a	7.6%	>	$1,000,000	7.6%

Pennsylvania		9.99%	>	$0	9.99%
Rhode Island		7%	>	$0	7%
South Carolina		5%	>	$0	5%
South Dakota			None		-
Tennessee	a	6.5%	>	$0	6.5%
Texas	a		(a)		0
Utah		4.95%	>	$0	4.95%
Vermont		6%	>	$0	7%
Vermont		7%	>	$10,000	7%
Vermont		8.5%	>	$25,000	7%
Virginia		6%	>	$0	6%
Washington	a		(a)		0
Washington, D.C.		8.25%	>	$0	8.25%
West Virginia		6.5%	>	$0	6.5%
Wisconsin		7.9%	>	$0	7.9%
Wyoming			None		-

(a) 2021 - Nevada, Ohio, Texas, and Washington do not have a corporate income tax but do have a gross receipts tax with rates not strictly comparable to corporate income tax rates. See Table 18 for more information. Delaware, Oregon, and Tennessee have gross receipts taxes in addition to corporate income taxes, as do several states like Pennsylvania, Virginia, and West Virginia, which permit gross receipts taxes at the local (but not state) level.

(b) 2021 - Florida's corporate income tax rate will return to 5.5% for tax years beginning on or after Jan. 1, 2022.

(c) 2021 - Georgia's corporate income tax rate will revert to 6% on January 1, 2026.

(d) 2021 - Illinois' rate includes two separate corporate income taxes, one at a 7% rate and one at a 2.5% rate.

(e) 2021 - Indiana's rate will change to 4.9% on July 1, 2021.

(f) 2021 - Mississippi continues to phase out the 3 percent bracket by increasing the exemption by $1,000 a year. By the start of 2022, the 3% bracket will be fully eliminated.

(g) 2021 - In New Jersey, the rates indicated apply to a corporation's entire net income rather than just income over the threshold. A temporary and retroactive surcharge is in effect from 2020 to 2023, bringing the rate to 11.5% for businesses with income over $1 million.

Note: In addition to regular income taxes, many states impose other taxes on corporations such as gross receipts taxes and capital stock taxes. Some states also impose an alternative minimum tax and special rates on financial institutions.

2022 Tax Brackets and Regulations

State	Category	Rates		Brackets	
Alabama		6.5%	>	$0	6.5%
Alaska		0.0%	>	$0	5.3%
Alaska		2.0%	>	$25,000	5.3%
Alaska		3.0%	>	$49,000	5.3%
Alaska		4.0%	>	$74,000	5.3%
Alaska		5.0%	>	$99,000	5.3%
Alaska		6.0%	>	$124,000	5.3%
Alaska		7.0%	>	$148,000	5.3%
Alaska		8.0%	>	$173,000	5.3%
Alaska		9.0%	>	$198,000	5.3%
Alaska		9.4%	>	$222,000	5.3%
Arizona		4.9%	>	$0	4.9%
Arkansas		1.0%	>	$0	3.4%
Arkansas		2.0%	>	$3,000	3.4%
Arkansas		3.0%	>	$6,000	3.4%
Arkansas		5.0%	>	$11,000	3.4%
Arkansas		5.9%	>	$25,000	3.4%
California		8.84%	>	$0	8.84%
Colorado		4.55%	>	$0	4.55%
Connecticut	a	7.50%	>	$0	7.50%
Delaware	b	8.7%	>	$0	8.7%
Florida		5.5%	>	$0	5.5%
Georgia	c	5.75%	>	$0	5.75%
Hawaii		4.4%	>	$0	5.4%
Hawaii		5.4%	>	$25,000	5.4%
Hawaii		6.4%	>	$100,000	5.4%
Idaho		6.0%	>	$0	6.0%
Illinois.	d	9.5%	>	$0	9.5%
Indiana		4.9%	>	$0	4.9%
Iowa		5.5%	>	$0	8.1%
Iowa		9.0%	>	$100,000	8.1%
Iowa		9.8%	>	$250,000	8.1%

Kansas		4.0%	>	$0	5.5%
Kansas		7.0%	>	$50,000	5.5%
Kentucky		5.0%	>	$0	5.0%
Louisiana		3.5%	>	$0	5.5%
Louisiana		5.5%	>	$50,000	5.5%
Louisiana		7.5%	>	$150,000	5.5%
Maine		3.50%	>	$0	8.40%
Maine		7.93%	>	$350,000	8.40%
Maine		8.33%	>	$1,050,000	8.40%
Maine		8.93%	>	$3,500,000	8.40%
Maryland		8.25%	>	$0	8.25%
Massachusetts		8.0%	>	$0	8.0%
Michigan		6.0%	>	$0	6.0%
Minnesota		9.8%	>	$0	9.8%
Mississippi	e	4.0%	>	$5,000	4.5%
Mississippi	e	5.0%	>	$10,000	4.5%
Missouri		4.0%	>	$0	4.0%
Montana		6.75%	>	$0	6.75%
Nebraska		5.58%	>	$0	6.54%
Nebraska		7.50%	>	$100,000	6.54%
Nevada	b		(b)		0
New Hampshire		7.6%	>	$0	7.6%
New Jersey	f	6.5%	>	$0	8.6%
New Jersey	f	7.5%	>	$50,000	8.6%
New Jersey	f	9.0%	>	$100,000	8.6%
New Jersey	f	11.5%	>	$1,000,000	8.6%
New Mexico		4.8%	>	$0	5.4%
New Mexico		5.9%	>	$500,000	5.4%
New York		6.50%	>	$0	6.88%
New York		7.25%	>	$5,000,000	6.88%
North Carolina		2.5%	>	$0	2.5%
North Dakota		1.41%	>	$0	3.09%
North Dakota		3.55%	>	$25,000	3.09%
North Dakota		4.31%	>	$50,000	3.09%
Ohio	b		(b)		0
Oklahoma		4.0%	>	$0	4.0%
Oregon	a	7.6%	>	$1,000,000	7.1%
Oregon	a	6.6%	>	$0	7.1%

Pennsylvania		9.99%	>	$0	9.99%
Rhode Island		7.0%	>	$0	7.0%
South Carolina		5.0%	>	$0	5.0%
South Dakota			None		-
Tennessee	b	6.5%	>	$0	6.5%
Texas	b		(b)		0
Utah		4.85%	>	$0	4.85%
Vermont		6.0%	>	$0	7.2%
Vermont		7.0%	>	$10,000	7.2%
Vermont		8.5%	>	$25,000	7.2%
Virginia		6.0%	>	$0	6.0%
Washington	b		(b)		0
Washington, D.C.		8.25%	>	$0	8.25%
West Virginia		6.5%	>	$0	6.5%
Wisconsin		7.9%	>	$0	7.9%
Wyoming			None		-

(a) 2022 - Connecticut charges a 10% surtax on a business's tax liability if it has gross proceeds of $100 million or more, or if it files as part of a combined unitary group. This surtax was recently extended and is scheduled to expire on January 1, 2023.
(b) 2022 - Nevada, Ohio, Texas, and Washington do not have a corporate income tax but do have a gross receipts tax with rates not strictly comparable to corporate income tax rates. Delaware, Oregon, and Tennessee have gross receipts taxes in addition to corporate income taxes, as do several states like Pennsylvania, Virginia, and West Virginia, which permit gross receipts taxes at the local (but not state) level.
(c) 2022 - Georgia's corporate income tax rate will revert to 6% on January 1, 2026.
(d) 2022 - Illinois' rate includes two separate corporate income taxes, one at a 7% rate and one at a 2.5% rate.
(e) 2022 - Mississippi finished phasing out its 3 percent bracket at the start of 2022.
(f) 2022 - In New Jersey, the rates indicated apply to a corporation's entire net income rather than just income over the threshold. A temporary and retroactive surcharge is in effect from 2020 to 2023, bringing the rate to 11.5% for businesses with income over $1 million.

2023 Tax Brackets and Regulations

State	Category	Rates		Brackets	Rate
Alabama		6.5%	>	$0	6.5%
Alaska		0.0%	>	$0	5.3%
Alaska		2.0%	>	$25,000	5.3%
Alaska		3.0%	>	$49,000	5.3%
Alaska		4.0%	>	$74,000	5.3%
Alaska		5.0%	>	$99,000	5.3%
Alaska		6.0%	>	$124,000	5.3%
Alaska		7.0%	>	$148,000	5.3%
Alaska		8.0%	>	$173,000	5.3%
Alaska		9.0%	>	$198,000	5.3%
Alaska		9.4%	>	$222,000	5.3%
Arizona		4.9%	>	$0	4.9%
Arkansas		1.0%	>	$0	3.3%
Arkansas		2.0%	>	$3,000	3.3%
Arkansas		3.0%	>	$6,000	3.3%
Arkansas		5.0%	>	$11,000	3.3%
Arkansas		5.3%	>	$25,000	3.3%
California		8.84%	>	$0	8.84%
Colorado		4.40%	>	$0	4.40%
Connecticut	a	7.5%	>	$0	7.5%
Delaware	b	8.7%	>	$0	8.7%
Florida		5.5%	>	$0	5.5%
Georgia	c	5.75%	>	$0	5.75%
Hawaii		4.4%	>	$0	5.4%
Hawaii		5.4%	>	$25,000	5.4%
Hawaii		6.4%	>	$100,000	5.4%
Idaho		5.8%	>	$0	5.8%
Illinois	d	9.5%	>	$0	9.5%
Indiana		4.9%	>	$0	4.9%
Iowa		5.5%	>	$0	7.0%
Iowa		8.4%	>	$100,000	7.0%
Kansas		4.0%	>	$0	5.5%
Kansas		7.0%	>	$50,000	5.5%
Kentucky		5.0%	>	$0	5.0%
Louisiana		3.5%	>	$0	5.5%
Louisiana		5.5%	>	$50,000	5.5%

Louisiana		7.5%	>	$150,000	5.5%
Maine		3.50%	>	$0	7.17%
Maine		7.93%	>	$350,000	7.17%
Maine		8.33%	>	$1,050,000	7.17%
Maine		8.93%	>	$3,500,000	7.17%
Maryland		8.25%	>	$0	8.25%
Massachusetts		8.0%	>	$0	8.0%
Michigan		6.0%	>	$0	6.0%
Minnesota		9.8%	>	$0	9.8%
Mississippi		4.0%	>	$5,000	4.5%
Mississippi		5.0%	>	$10,000	4.5%
Missouri		4.0%	>	$0	4.0%
Montana		6.75%	>	$0	6.75%
Nebraska		5.58%	>	$0	6.42%
Nebraska		7.25%	>	$100,000	6.42%
Nevada	b		(b)		0
New Hampshire		7.5%	>	$0	7.5%
New Jersey	e	6.5%	>	$0	8.6%
New Jersey	e	7.5%	>	$50,000	8.6%
New Jersey	e	9.0%	>	$100,000	8.6%
New Jersey	e	11.5%	>	$1,000,000	8.6%
New Mexico		4.8%	>	$0	5.4%
New Mexico		5.9%	>	$500,000	5.4%
New York		6.50%	>	$0	6.88%
New York		7.25%	>	$5,000,000	6.88%
North Carolina		2.5%	>	$0	2.5%
North Dakota		1.41%	>	$0	3.09%
North Dakota		3.55%	>	$25,000	3.09%
North Dakota		4.31%	>	$50,000	3.09%
Ohio	b		(b)		0
Oklahoma		4.0%	>	$0	4.0%
Oregon	a	6.6%	>	$0	7.1%
Oregon	a	7.6%	>	$1,000,000	7.1%
Pennsylvania		8.99%	>	$0	8.99%
Rhode Island		7.0%	>	$0	7.0%
South Carolina		5.0%	>	$0	5.0%
South Dakota			None		-
Tennessee	b	6.5%	>	$0	6.5%

Texas	b		(b)		0
Utah		4.85%	>	$0	4.85%
Vermont		6.0%	>	$0	7.2%
Vermont		7.0%	>	$10,000	7.2%
Vermont		8.5%	>	$25,000	7.2%
Virginia		6.0%	>	$0	6.0%
Washington	b		(b)		0
Washington, D.C.		8.25%	>	$0	8.25%
West Virginia		6.5%	>	$0	6.5%
Wisconsin		7.9%	>	$0	7.9%
Wyoming			None		-

(a) 2023 - Connecticut has historically charged a 10% surtax on a business's tax liability if it has gross proceeds of $100 million or more, or if it files as part of a combined unitary group. This surtax expired on January 1. Legislators have extended the surtax in the past and will decide whether to do so again this session.
(b) 2023 - Nevada, Ohio, Texas, and Washington do not have a corporate income tax but do have a gross receipts tax with rates not strictly comparable to corporate income tax rates. Delaware, Oregon, and Tennessee have gross receipts taxes in addition to corporate income taxes, as do several states like Pennsylvania, Virginia, and West Virginia, which permit gross receipts taxes at the local (but not state) level.
(c) 2023 - Georgia's corporate income tax rate will revert to 6% on January 1, 2026.
(d) 2023 - Illinois' rate includes two separate corporate income taxes, one at a 7% rate and one at a 2.5% rate.
(e) 2023 - In New Jersey, the rates indicated apply to a corporation's entire net income rather than just income over the threshold. A temporary and retroactive surcharge is in effect from 2020 to 2023, bringing the rate to 11.5% for businesses with income over $1 million.
Note: In addition to regular income taxes, many states impose other taxes on corporations such as gross receipts taxes and capital stock taxes. Some states also impose an alternative minimum tax and special rates on financial institutions.

Top State to Start a Business Rankings:

2021:
https://www.cnbc.com/2021/07/13/americas-top-states-for-business.html

2022:
https://www.cnbc.com/2022/07/13/americas-top-states-for-business-2022-the-full-rankings.html

2023:
https://www.cnbc.com/2023/07/11/americas-top-states-for-business-2023-the-full-rankings.html

Regression Analysis for Top State to Start a Business in 2021

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.982763275
R Square	0.965823656
Adjusted R Square	0.95706049
Standard Error	3.020702864
Observations	50

ANOVA

	df	*SS*	*MS*	*F*	***Significance F***
Regression	10	10056.63881	1005.66388	110.214019	**2.17508E-25**
Residual	39	355.8611859	9.12464579		
Total	49	10412.5			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	-45.01045788	3.013232625	-14.937598	1.0032E-17	-51.10529615	-38.91561961	-51.105296	-38.91562
COST OF DOING BUSINESS	0.231439973	0.064236188	3.60295311	0.00087913	0.101510019	0.361369927	0.10151002	0.36136993
INFRASTRUCTURE	0.309591428	0.042028658	7.36619834	6.7691E-09	0.224580444	0.394602412	0.22458044	0.39460241
LIFE, HEALTH & INCLUSION	0.508420268	0.054516058	9.32606436	1.7767E-11	0.398151132	0.618689404	0.39815113	0.6186894
WORKFORCE	0.223374638	0.051405984	4.34530424	9.624E-05				
ECONOMY	0.306331076	0.039899032	7.6776569	2.5622E-09				
BUSINESS FRIENDLINESS	0.241982322	0.042587347	5.68202386	1.4317E-06				
ACCESS TO CAPITAL	0.346508509	0.072279232	4.79402586	2.3937E-05				
TECHNOLOGY & INNOVATION	0.190723078	0.05296807	3.60071789	0.00088481				
EDUCATION	0.07053192	0.048833411	1.44433737	0.15663059				
COST OF LIVING	0.353286065	0.060897261	5.80134574	9.7768E-07				

0.119396222	0.327353055	0.11939622	0.32735305
0.225627667	0.387034485	0.22562767	0.38703449
0.155841282	0.328123362	0.15584128	0.32812336
0.200309962	0.492707056	0.20030996	0.49270706
0.083585043	0.297861112	0.08358504	0.29786111
-0.028242977	0.169306816	-0.028243	0.16930682
0.230109728	0.476462402	0.23010973	0.4764624

RESIDUAL OUTPUT

Observation	*Predicted OVERALL*	*Residuals*
1	-1.039604165	2.039604165
2	2.521445629	-0.521445629
3	3.645544789	-0.645544789
4	5.988162515	-1.988162515
5	7.906214772	-2.906214772
6	7.566404003	-1.566404003
7	5.791381083	1.208618917
8	8.750530977	-0.750530977
9	10.74750794	-1.747507941
10	12.31416936	-2.31416936
11	13.07998463	-2.07998463
12	15.67126675	-3.671266755
13	14.86661189	-1.866611893
14	16.48887866	-2.488878663
15	16.41939253	-1.419392533
16	19.08370696	-3.083706957
17	18.1530329	-1.153032899
18	19.61855229	-1.618552294
19	16.94464797	2.05535203
20	18.34393992	1.656060083
21	25.628325	-4.628324999
22	17.71685648	4.283143517
23	28.02287043	-5.02287043
24	21.99823533	2.00176467
25	21.98171678	3.018283215
26	22.44674507	3.553254933
27	23.95114777	3.048852227
28	24.09141546	3.908584544
29	27.91786824	1.082131762

30	26.55237151	3.44762849
31	33.13239263	-2.132392629
32	30.33185598	1.668144023
33	28.7427819	4.257218095
34	31.41554526	2.584454741
35	32.98650774	2.013492258
36	35.40538216	0.594617837
37	34.08921932	2.910780682
38	35.85120572	2.148794278
39	37.25106126	1.748938744
40	40.22328023	-0.223280232
41	38.16521431	2.834785686
42	43.73481562	-1.734815624
43	41.32808542	1.67191458
44	46.97211492	-2.972114915
45	43.24428995	1.75571005
46	45.03309295	0.966907047
47	49.51409265	-2.514092649
48	47.31819039	0.681809612
49	49.27206464	-0.272064636
50	57.81948346	-7.819483459

Regression Analysis for Top State to Start a Business in 2022

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.975241
R Square	0.951096
Adjusted R Square	0.938556
Standard Error	3.597387
Observations	50

ANOVA

	df	*SS*	*MS*	*F*
Regression	10	9815.613508	981.561351	75.84783088
Residual	39	504.7064924	12.9411921	
Total	49	10320.32		

Significance F
2.22E-22

	Coefficients	*Standard Error*	*t Stat*	*P-value*
Intercept	-35.1381	2.863623176	-12.270492	5.75988E-15
WORKFORCE	0.25439	0.04790195	5.31063561	4.67998E-06
INFRA-STRUCTURE	0.22817	0.0500703	4.55698352	5.01159E-05
COST OF DOING BUSINESS	0.377646	0.069200522	5.45726921	2.93402E-06
ECONOMY	0.396651	0.046752218	8.48410573	2.16548E-10
LIFE, HEALTH & INCLUSION	0.466413	0.051438151	9.06745673	3.7947E-11
TECHNOLOGY & INNOVATION	0.236093	0.078067214	3.02422509	0.004393625
BUSINESS FRIENDLINESS	0.093507	0.046483818	2.01161139	0.051206982
EDUCATION	0.109069	0.054263801	2.00998312	0.051386481
ACCESS TO CAPITAL	0.064465	0.055618358	1.15906768	0.253476361
COST OF LIVING	0.16498	0.062892784	2.62318758	0.012366406

Lower 95%	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
-40.9303	-29.34584189	-40.930291	-29.34584189
0.157499	0.351280637	0.15749896	0.351280637

0.126893	0.329446275	0.12689279	0.329446275
0.237675	0.517617145	0.23767461	0.517617145
0.302085	0.491216045	0.30208547	0.491216045
0.36237	0.570456691	0.36236973	0.570456691
0.078187	0.393998671	0.07818698	0.393998671
-0.00052	0.187529773	-0.000515	0.187529773
-0.00069	0.218828222	-0.0006896	0.218828222
-0.04803	0.176964189	-0.0480333	0.176964189
0.037767	0.292192235	0.03776691	0.292192235

RESIDUAL OUTPUT

Observation	*Predicted OVERALL RANK*	*Residuals*
1	3.591585	-2.591585003
2	1.73078	0.269219991
3	3.280222	-0.280221662
4	4.170665	-0.170665431
5	8.041479	-3.04147869
6	6.111129	-0.111129035
7	7.089309	-0.089309261
8	12.96775	-4.967748609
9	6.451109	2.548890962
10	11.86619	-1.86619384
11	15.82985	-4.829848964
12	9.484365	2.515635133
13	19.35577	-6.355765226
14	15.33194	-1.331941754
15	12.68491	2.315093989
16	12.11104	3.888962294
17	21.63019	-4.630189779
18	17.32799	0.672010647
19	22.1465	-3.14650487
20	22.44605	-2.446049466
21	22.27978	-1.279776457
22	24.45203	-2.452034829
23	26.36279	-3.362788702
24	25.49597	-1.495968627
25	21.84072	3.159278355
26	24.46588	1.534116586

27	25.811	1.188995691
28	25.93479	2.065206511
29	25.54091	3.459088595
30	30.48085	-0.480847736
31	31.46743	-0.467430338
32	30.30131	1.698692262
33	28.02697	4.973028834
34	33.1653	0.834695808
35	31.5749	3.425104163
36	35.38413	0.615869962
37	34.29705	1.702946604
38	33.57334	4.42666406
39	37.46136	1.538643402
40	34.62698	4.373019418
41	32.19023	8.809768265
42	36.80707	5.192926087
43	39.84989	3.150114268
44	45.20669	-1.206686286
45	49.41622	-4.416222815
46	46.18361	-0.183612103
47	51.88583	-5.88582787
48	50.52485	-2.524851046
49	51.34866	-2.34866183
50	52.39463	-2.394631659

Regression Analysis for Top State to Start a Business in 2023

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.96928072
R Square	0.93950512
Adjusted R Square	0.92399361
Standard Error	4.02348868
Observations	50

ANOVA

	df	*SS*	*MS*	*F*
Regression	10	9805.07002	980.507002	60.5682648
Residual	39	631.349984	16.1884611	
Total	49	10436.42		

Significance F
1.3422E-20

	Coefficients	*Standard Error*	*t Stat*	*P-value*
Intercept	-31.087123	4.09067489	-7.5995097	3.2668E-09
WORK- FORCE	0.28535975	0.05823998	4.89972272	1.7178E-05
INFRA- STRUCTURE	0.2309781	0.06235953	3.70397437	0.00065615
ECONOMY	0.3457802	0.07196989	4.80451189	2.3163E-05
LIFE, HEALTH & INCLUSION	0.33189194	0.07433002	4.46511308	6.6584E-05
COST OF DOING BUSINESS	0.29128857	0.08857618	3.28856555	0.00213922
TECHN & INNOVATION	0.36197191	0.07331092	4.93748964	1.5252E-05
BUSINESS FRIENDLI- NESS	0.16641482	0.04384879	3.79519754	0.00050253
EDUCATION	0.06196004	0.04476747	1.38404158	0.1742155
ACCESS TO CAPITAL	0.05465578	0.06688901	0.81711158	0.41882845
COST OF LIVING	0.10227216	0.07429373	1.3765921	0.17649055

Lower 95%	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
-39.361294	-22.812953	-39.361294	-22.812953
0.16755827	0.40316123	0.16755827	0.40316123
0.10484404	0.35711215	0.10484404	0.35711215
0.20020736	0.49135305	0.20020736	0.49135305
0.18154528	0.48223859	0.18154528	0.48223859

0.11212634	0.47045081	0.11212634	0.47045081
0.21368658	0.51025724	0.21368658	0.51025724
0.07772227	0.25510737	0.07772227	0.25510737
-0.0285907	0.15251081	-0.0285907	0.15251081
-0.08064	0.18995157	-0.08064	0.18995157
-0.0480011	0.25254542	-0.0480011	0.25254542

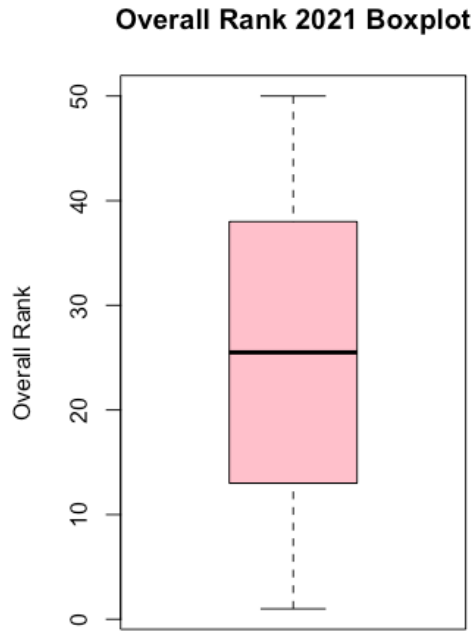
RESIDUAL OUTPUT

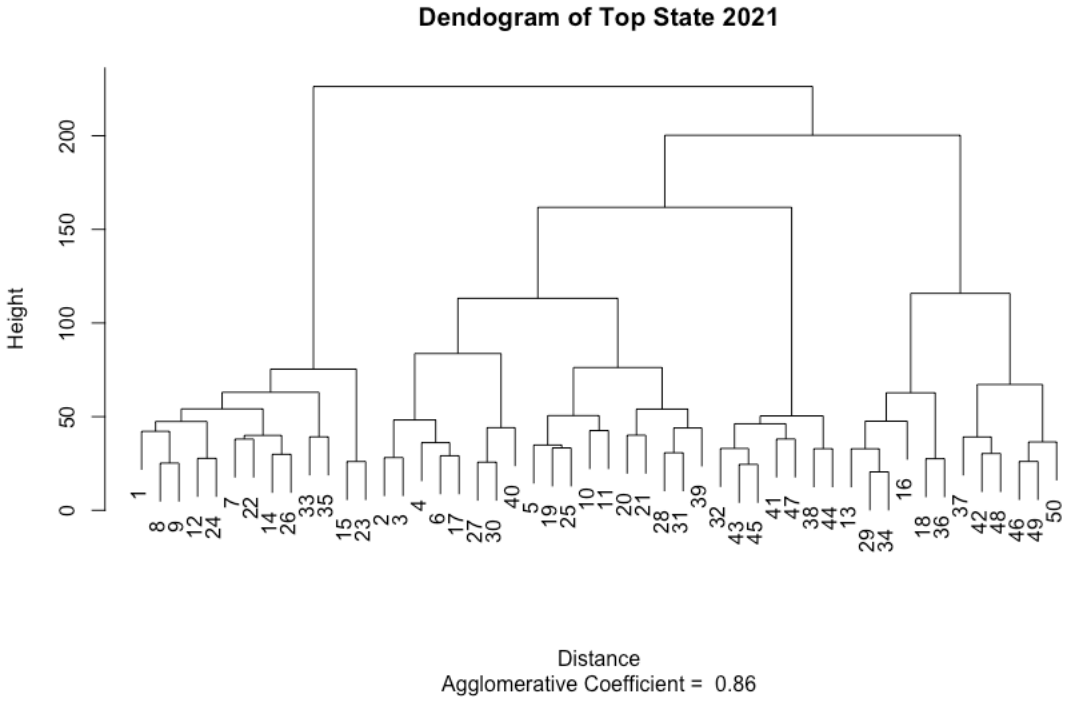
Observation	*Predicted OVERALL RANK*	*Residuals*
1	-2.1822704	3.18227036
2	2.19485567	-0.1948557
3	6.52807056	-3.5280706
4	10.3104696	-6.3104696
5	5.52102872	-0.5210287
6	8.50323545	-2.5032355
7	6.7902423	0.2097577
8	16.3068941	-8.3068941
9	15.8805202	-6.8805202
10	11.8406147	-1.8406147
11	10.7317049	0.26829514
12	12.994469	-0.994469
13	14.65712	-1.65712
14	14.1024899	-0.1024899
15	15.074862	-0.074862
16	13.7745102	1.22548981
17	16.6691387	0.33086135
18	17.5126963	0.4873037
19	18.2179095	0.78209049
20	22.2385878	-2.2385878
21	18.58536	2.41464004
22	19.7289751	2.27102493
23	25.0358328	-2.0358328
24	24.4696305	-1.4696305
25	24.3575247	0.64247532
26	23.2547621	2.74523788
27	25.8608541	1.13914586
28	29.7917832	-1.7917832

29	34.5618284	-5.5618284
30	23.6181339	6.38186606
31	26.0215172	4.97848278
32	26.808043	5.19195702
33	28.9894001	4.01059988
34	34.6519594	-0.6519594
35	34.7746686	0.22533137
36	38.6589648	-2.6589648
37	33.9228959	3.07710412
38	36.4597533	1.54024666
39	36.3260283	2.6739717
40	38.6655129	1.33448714
41	33.4168476	7.58315239
42	34.3217199	7.67828012
43	39.700036	3.29996404
44	41.5153132	2.48468676
45	42.206904	2.79309601
46	47.9615585	-1.9615585
47	50.8449982	-3.8449982
48	54.4637729	-6.4637729
49	54.2507602	-5.2507602
50	52.107512	-2.107512

Hierarchical Clustering for Top State to Start a Business in 2021

```
#2021
X2021_best_state <- read_excel("Desktop/Mercy /Winter 2023/801 Capstone/Thesis /2021 best state.xlsx",  + sheet = "Sheet1", col_types = c("numeric",
                                                                                                                                    +         "text", "numeric", "numeric", "numeric",
                                                                                                                                    +         "numeric", "numeric", "numeric",
                                                                                                                                    +         "numeric", "numeric", "numeric",
                                                                                                                                    +         "numeric", "skip", "skip", "skip",
                                                                                                                                    +         "skip", "skip"))
View(X2021_best_state)
str(X2021_best_state)
top21 <- X2021_best_state
fivenum(top21$OVERALL)
fivenum(top21$WORKFORCE)
fivenum(top21$`INFRA-STRUCTURE`)
fivenum(top21$`COST OF DOING BUSINESS`)
fivenum(top21$ECONOMY)
fivenum(top21$`LIFE, HEALTH & INCLUSION`)
fivenum(top21$`TECHNOLOGY & INNOVATION`)
fivenum(top21$`BUSINESS FRIENDLINESS`)
fivenum(top21$EDUCATION)
fivenum(top21$`ACCESS TO CAPITAL`)
fivenum(top21$`COST OF LIVING`)
summary(top21$OVERALL)
boxplot(top21$OVERALL,col = "pink", main="Overall Rank 2021 Boxplot", ylab="Overall Rank"
#2021 Hierarchical Cluster
install.packages("cluster")
library(cluster)
top21.scale <- top21[,2:12]
d21 <- dist(top21.scale, method = "euclidean")
top21result <- agnes(d21,diss=TRUE,method = "ward")
top21result
#agglomerative coefficient: 0.86 --> good natural clustering structure
plot(top21result,main = "Dendogram of Top State 2021", xlab="Distance")
top21clusters <- cutree(top21result, k=3)
top21<- data.frame(top21, top21clusters)
View(top21)
summary(subset(top21, top21clusters==1))
summary(subset(top21, top21clusters==3))
```


Overall Rank 2021 Boxplot
Overall Rank
0
10
20
30
40
50


Dendogram of Top State 2021
Height
0
50
100
150
200
1
8
9
12
24
7
22
14
26
33
35
15
23
2
3
4
6
17
27
30
40
5
19
25
10
11
20
21
28
31
39
32
43
45
41
47
38
44
13
29
34
16
18
36
37
42
48
46
49
50
Distance
Agglomerative Coefficient = 0.86

```
> summary(subset(top21, top21clusters==3))
    OVERALL         STATE.
 Min.   :13.00   Length:12
 1st Qu.:26.25   Class :character
 Median :36.50   Mode  :character
 Mean   :34.83
 3rd Qu.:46.50
 Max.   :50.00
 COST.OF.DOING.BUSINESS INFRASTRUCTURE
 Min.   :12.00          Min.   :17.00
 1st Qu.:19.50          1st Qu.:36.00
 Median :30.00          Median :42.50
 Mean   :30.75          Mean   :39.83
 3rd Qu.:41.50          3rd Qu.:46.50
 Max.   :50.00          Max.   :50.00
 LIFE..HEALTH...INCLUSION   WORKFORCE
 Min.   : 1.0             Min.   :18.00
 1st Qu.: 4.5             1st Qu.:29.25
 Median :12.0             Median :36.00
 Mean   :11.5             Mean   :36.17
 3rd Qu.:17.5             3rd Qu.:47.25
 Max.   :22.0             Max.   :50.00
    ECONOMY      BUSINESS.FRIENDLINESS
 Min.   : 1.00   Min.   : 1.00
 1st Qu.:17.75   1st Qu.: 3.75
 Median :35.00   Median : 9.50
 Mean   :29.92   Mean   :14.00
 3rd Qu.:38.50   3rd Qu.:23.25
 Max.   :49.00   Max.   :36.00
 ACCESS.TO.CAPITAL TECHNOLOGY...INNOVATION
 Min.   :34.00     Min.   :18.00
 1st Qu.:40.75     1st Qu.:30.25
 Median :41.00     Median :38.00
 Mean   :42.75     Mean   :35.42
 3rd Qu.:46.25     3rd Qu.:41.25
 Max.   :49.00     Max.   :46.00
   EDUCATION     COST.OF.LIVING  top21clusters
 Min.   : 4.00   Min.   :16.00   Min.   :3
 1st Qu.:12.50   1st Qu.:23.75   1st Qu.:3
 Median :21.00   Median :32.00   Median :3
 Mean   :22.50   Mean   :32.25   Mean   :3
 3rd Qu.:28.75   3rd Qu.:40.50   3rd Qu.:3
 Max.   :47.00   Max.   :50.00   Max.   :3
```

Hierarchical Clustering for Top State to Start a Business in 2022

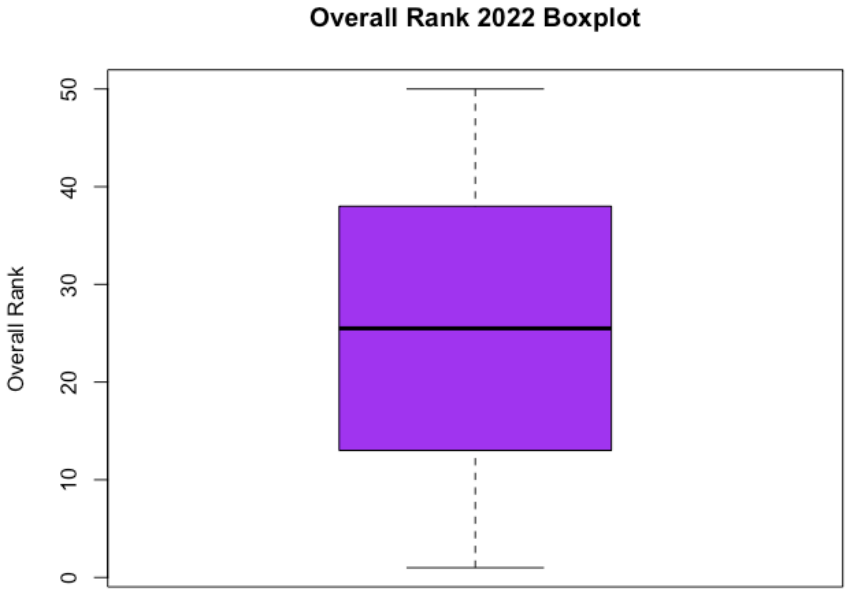
#2022

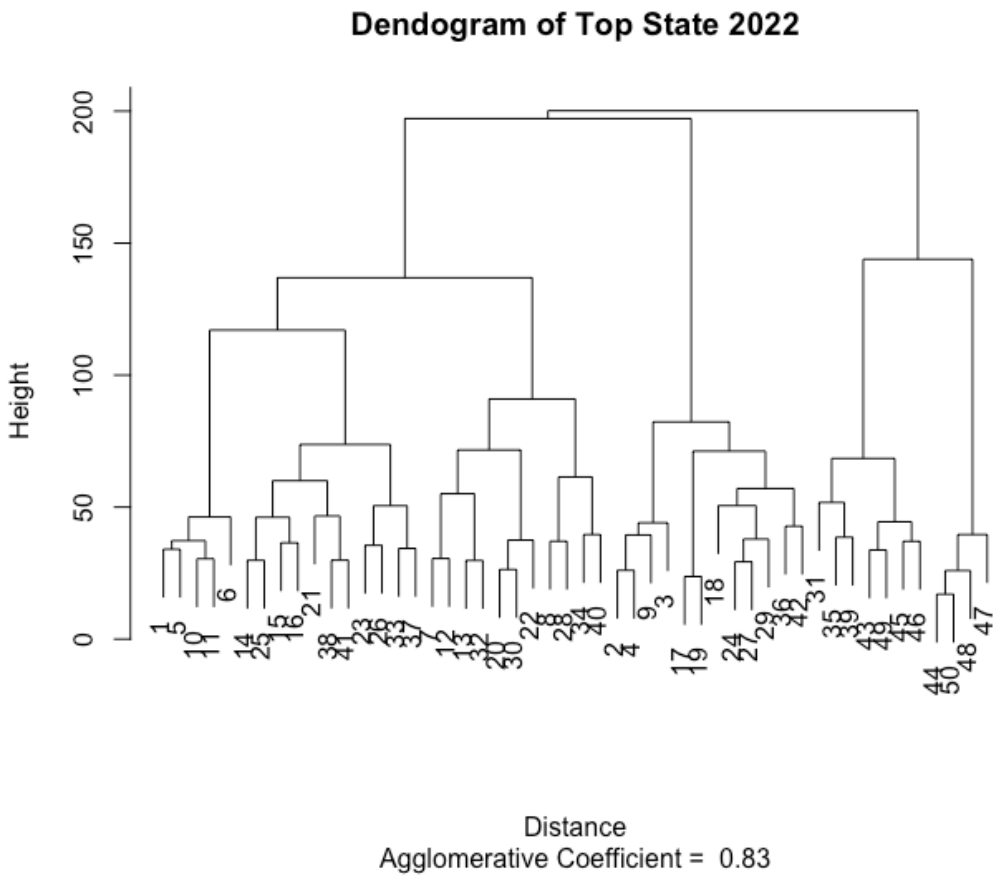
```
library(readxl)
X2022_top_states_for_businesses <- read_excel("Desktop/Mercy /Winter 2023/801 Capstone/Thesis /2022 top states for businesses.xlsx", col_types = c("numeric", "text", "numeric", "numeric", "numeric", "numeric",  "numeric", "numeric", "numeric", "numeric", "numeric", "numeric", "skip"))
View(X2022_top_states_for_businesses)

install.packages("ggplot2")
library(ggplot2)
top22 <- X2022_top_states_for_businesses
str(top22)
fivenum(top22$`OVERALL RANK `)
fivenum(top22$WORKFORCE)
fivenum(top22$`INFRA-STRUCTURE`)
fivenum(top22$`COST OF DOING BUSINESS`)
fivenum(top22$ECONOMY)
fivenum(top22$`LIFE, HEALTH & INCLUSION`)
fivenum(top22$`TECHNOLOGY & INNOVATION`)
fivenum(top22$`BUSINESS FRIENDLINESS`)
fivenum(top22$EDUCATION)
fivenum(top22$`ACCESS TO CAPITAL`)
fivenum(top22$`COST OF LIVING`)

summary(top22$`OVERALL RANK `)
boxplot(top22$`OVERALL RANK `,col = "purple", main="Overall Rank 2022 Boxplot", ylab="Overall Rank")
#2022 Hierarchical Cluster
top22.scale <- top22[,2:12]
d22 <- dist(top22.scale, method = "euclidean")
top22result <- agnes(d22,diss=TRUE,method = "ward")
top22result
#agglomerative coefficient: 0. --> good natural clustering structure
plot(top22result,main = "Dendogram of Top State 2022", xlab="Distance")
top22clusters <- cutree(top22result, k=3)
top22<- data.frame(top22, top22clusters)
View(top22)
summary(subset(top22, top22clusters==1))
```

summary(subset(top22, top21clusters==3))





```
> summary(subset(top22, top22clusters==3))
 OVERALL.RANK.      STATE.             WORKFORCE     INFRA.STRUCTURE
 Min.   :31.00   Length:11          Min.   :14.00   Min.   :22.00
 1st Qu.:41.00   Class :character   1st Qu.:23.50   1st Qu.:39.00
 Median :45.00   Mode  :character   Median :39.00   Median :44.00
 Mean   :43.27                      Mean   :34.36   Mean   :42.27
 3rd Qu.:47.00                      3rd Qu.:43.00   3rd Qu.:47.50
 Max.   :50.00                      Max.   :50.00   Max.   :50.00

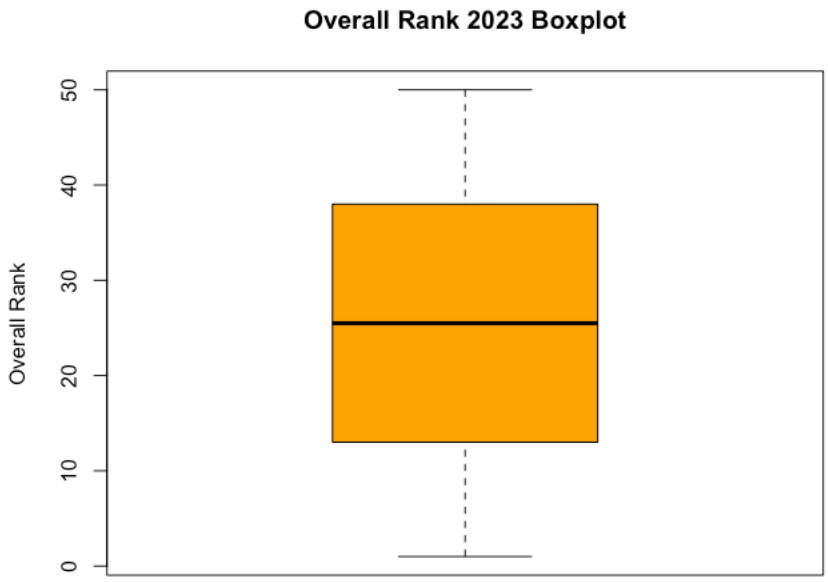
 COST.OF.DOING.BUSINESS    ECONOMY      LIFE..HEALTH...INCLUSION
 Min.   : 5.00          Min.   :29.00   Min.   : 1.00
 1st Qu.:23.00          1st Qu.:35.50   1st Qu.: 7.00
 Median :39.00          Median :41.00   Median :16.00
 Mean   :32.73          Mean   :40.27   Mean   :20.09
 3rd Qu.:45.50          3rd Qu.:46.00   3rd Qu.:33.50
 Max.   :50.00          Max.   :49.00   Max.   :45.00

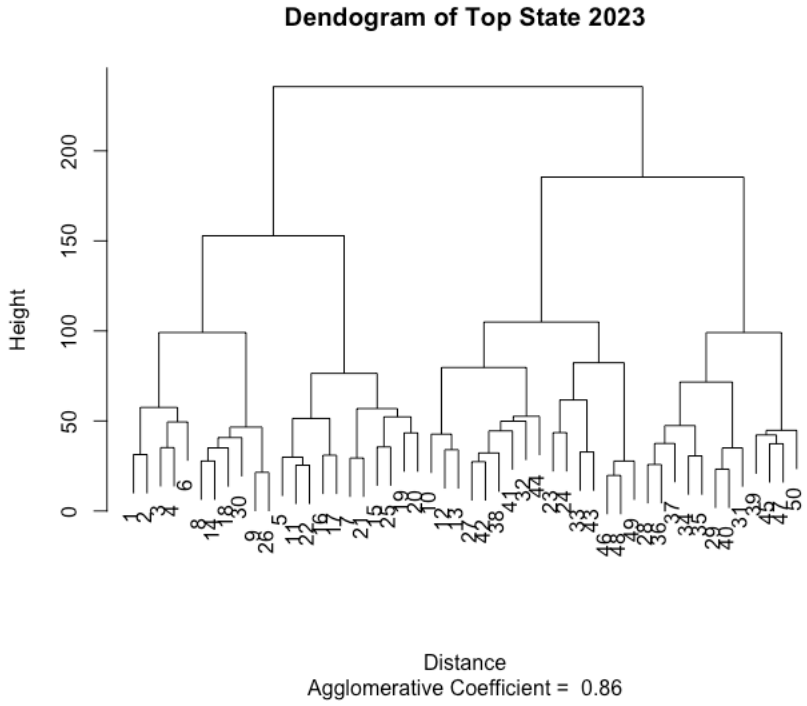
 TECHNOLOGY...INNOVATION BUSINESS.FRIENDLINESS   EDUCATION     ACCESS.TO.CAPITAL
 Min.   :25.00           Min.   : 8.0          Min.   : 6.00   Min.   :26.00
 1st Qu.:38.00           1st Qu.:13.0          1st Qu.:11.75   1st Qu.:37.25
 Median :43.00           Median :35.0          Median :32.00   Median :43.00
 Mean   :41.18           Mean   :29.0          Mean   :28.50   Mean   :40.90
 3rd Qu.:46.50           3rd Qu.:42.5          3rd Qu.:42.50   3rd Qu.:47.75
 Max.   :50.00           Max.   :50.0          Max.   :49.00   Max.   :49.00
                                               NA's   :1       NA's   :1
 COST.OF.LIVING  top22clusters
 Min.   : 8.00   Min.   :3
 1st Qu.:21.25   1st Qu.:3
 Median :40.00   Median :3
 Mean   :33.20   Mean   :3
 3rd Qu.:42.75   3rd Qu.:3
 Max.   :50.00   Max.   :3
 NA's   :1
```

Hierarchical Clustering for Top State to Start a Business in 2023

```
#2023
X2023_top_states_for_business <- read_excel("Desktop/Mercy /Winter 2023/801
Capstone/Thesis /2023 top states for business.xlsx",
                              +     col_types = c("numeric", "text", "numeric",
                                                  +         "numeric", "numeric", "numeric",
                                                  +         "numeric", "numeric", "numeric",
                                                  +         "numeric", "numeric", "numeric",
                                                  +         "skip"))
View(X2023_top_states_for_business)
top23 <- X2023_top_states_for_business
str(top23)
fivenum(top23$`OVERALL RANK `)
fivenum(top23$WORKFORCE)
fivenum(top23$`INFRA-STRUCTURE`)
fivenum(top23$`COST OF DOING BUSINESS`)
fivenum(top23$ECONOMY)
fivenum(top23$`LIFE, HEALTH & INCLUSION`)
fivenum(top23$`TECHNOLOGY & INNOVATION`)
fivenum(top23$`BUSINESS FRIENDLINESS`)
fivenum(top23$EDUCATION)
fivenum(top23$`ACCESS TO CAPITAL`)
fivenum(top23$`COST OF LIVING`)
summary(top23$`OVERALL RANK `)

boxplot(top21$OVERALL,col = "orange", main="Overall Rank 2023 Boxplot", ylab="Overall
Rank")
#2023 Hierarchical Cluster
top23.scale <- top23[,2:12]
d23 <- dist(top23.scale, method = "euclidean")
top23result <- agnes(d23,diss=TRUE,method = "ward")
top23result
#agglomerative coefficient: 0. --> good natural clustering structure
plot(top23result,main = "Dendogram of Top State 2023", xlab="Distance")
top23clusters <- cutree(top23result, k=3)
top23<- data.frame(top23, top23clusters)
View(top23)
summary(subset(top23, top23clusters==1))
summary(subset(top23, top23clusters==3))
```


Overall Rank 2023 Boxplot
Overall Rank
0
10
20
30
40
50


Dendogram of Top State 2023
Height
0
50
100
150
200
Distance
Agglomerative Coefficient = 0.86

```
> summary(subset(top23, top23clusters==3))
 OVERALL RANK       STATE              WORK- FORCE    INFRA- STRUCTURE    ECONOMY
 Min.   :28.00   Length:12          Min.   :18.00   Min.   :25.00    Min.   :23.00
 1st Qu.:33.25   Class :character   1st Qu.:21.75   1st Qu.:34.50    1st Qu.:34.75
 Median :36.50   Mode  :character   Median :35.00   Median :38.50    Median :40.00
 Mean   :37.58                      Mean   :32.42   Mean   :38.75    Mean   :39.00
 3rd Qu.:41.25                      3rd Qu.:41.25   3rd Qu.:43.25    3rd Qu.:44.50
 Max.   :50.00                      Max.   :47.00   Max.   :50.00    Max.   :50.00
 LIFE, HEALTH & INCLUSION COST OF DOING BUSINESS TECHN & INNOVATION
 Min.   : 1.00            Min.   :11.00          Min.   :13.00
 1st Qu.: 8.75            1st Qu.:26.50          1st Qu.:32.50
 Median :16.50            Median :35.00          Median :37.00
 Mean   :15.75            Mean   :33.83          Mean   :35.83
 3rd Qu.:22.75            3rd Qu.:43.75          3rd Qu.:42.25
 Max.   :31.00            Max.   :50.00          Max.   :50.00
 BUSINESS FRIENDLI- NESS    EDUCATION     ACCESS TO CAPITAL COST OF LIVING
 Min.   : 1.00            Min.   : 8.00   Min.   :27.00     Min.   :12.00
 1st Qu.: 3.50            1st Qu.:12.50   1st Qu.:31.50     1st Qu.:25.75
 Median :10.00            Median :19.50   Median :38.50     Median :33.00
 Mean   :13.67            Mean   :22.33   Mean   :38.00     Mean   :32.50
 3rd Qu.:18.50            3rd Qu.:29.25   3rd Qu.:42.75     3rd Qu.:40.75
 Max.   :37.00            Max.   :49.00   Max.   :50.00     Max.   :50.00
```

Boxplot for Top State to Start a Business from 2021-2023 for Variable ‘Overall Ranking’

boxplot(top22$`OVERALL RANK `, top21$OVERALL,top23$`OVERALL RANK `, col = "pink", main="Overall Best State Rank from 2021-2023 Boxplot", ylab="Overall Rank")

